

2026S1 PH1012: Physics A
Basics and Fluids

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Week 1
Serway Chap 1, 14.1 – 14.4

"The true sign of intelligence is not knowledge but imagination."
- Albert Einstein (1879-1955)

Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. perform units conversion, work with common prefixes
2. analyze equations to evaluate their homogeneity
3. make estimates of physical quantities
4. analyze and solve problems involving mass, weight and density
5. identify basic forces in simple systems
6. analyze and solve problems involving pressure (solid / liquids)
7. apply Archimedes principle and solve problems involving upthrust (buoyant forces)
8. interpret and evaluate basic scientific / technical content in public talks / articles

Overall View of Topics

Basic Fluids

Kinematics

Circular Motion

Newton's Laws and Forces

Work done and Energy

Thermal Physics

Gas Laws and
Thermodynamics

Rotational Dynamics

Kinematics, Fluids



https://commons.wikimedia.org/wiki/File:Marina_Bay_Sands_in_the_evening_-_20101120.jpg

Basic Structural Mechanics;
Physics and Environment

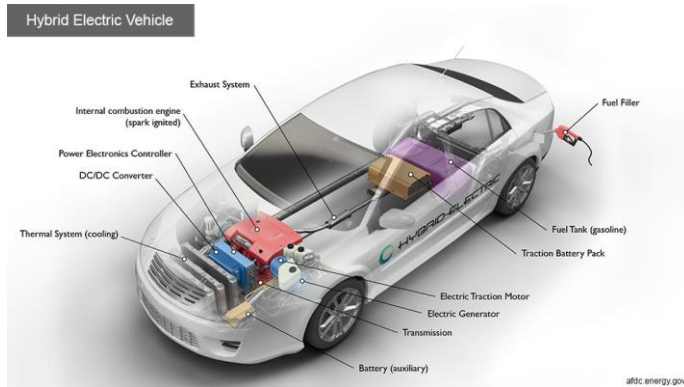


<https://blogs.icrc.org/law-and-policy/2022/03/15/consumer-drones-conflict-ihl/>

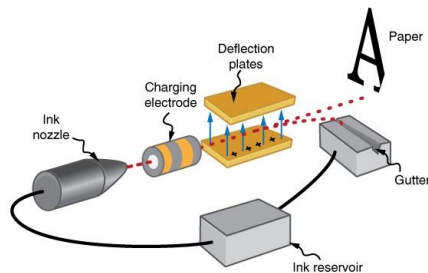
Circuits, Charge

Electromagnetism - Motors

Basic Materials



<https://www.accessscience.com/binary/mheasworks/72fbd6ef9c9e7775/7fe79ef815f5ec3190d73229d5577e0e461d8c5997bbe39282d6c41c019e21d6/326340FG0010.jpg>



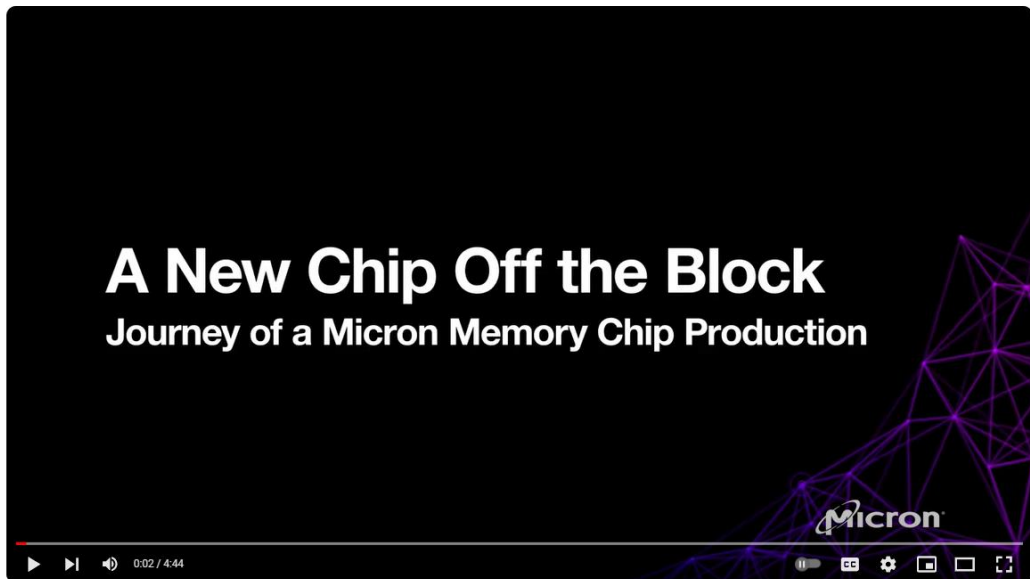
<https://openstax.org/books/college-physics/pages/18-8-applications-of-electrostatics>

Charge, Electric Fields



Which aspects of Physics do you need
to analyze and control a 3D printer?

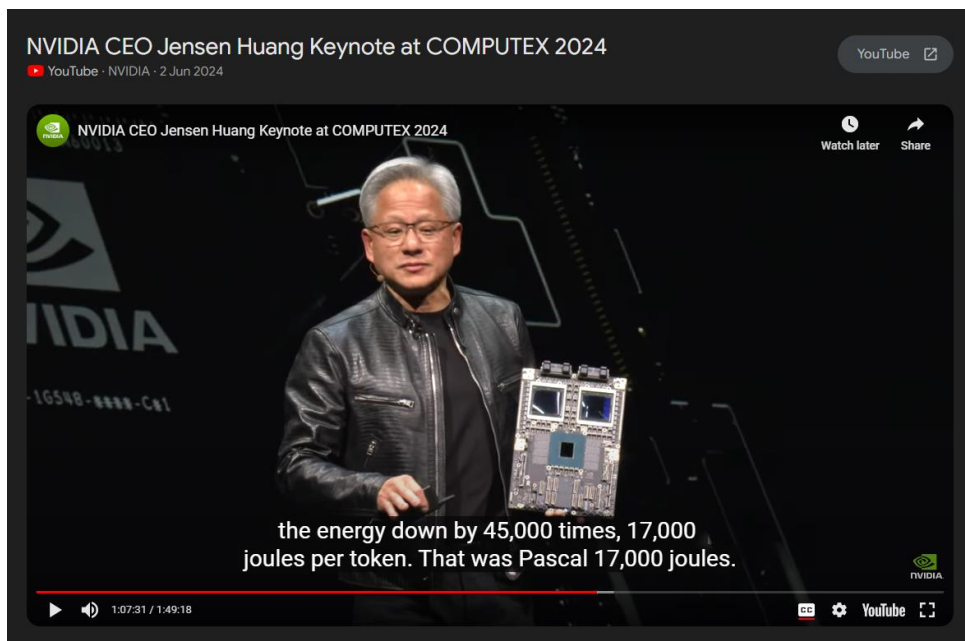
Watch this Micron Video... What expertise are needed to make this technology become a reality?



A New Chip Off the Block: Journey of a Memory Chip Production | Micron Singapore

<https://www.youtube.com/watch?v=qllBM1NP3U4&t=2s>

Based on your knowledge of units and Physics, are you able to follow Jensen Huang's Keynote at COMPUTEX 2024?



<https://www.youtube.com/watch?v=pKXDVsWZmUU>

0:59:10 – 1:00:40 AI and Physics

1:05:00 – 1:15:00 Energy Discussion

1:29:10 – 1:45:00 Physics, AI and Omniverse

Class Discussion: Video

Units

Units and Measurements (Pre-lecture Video)

The *Système International d'Unités* (SI), or International System of Units, defines seven units of measure as a basic set from which all other SI units are derived. The SI base units and their physical quantities are:

Base quantity	SI Base Unit	Symbol	Example
length	meter	m	Human height about 1.5 – 1.8 m
mass	kilogram	kg	Mass of 1 litre of water: 1 kg
time	second	s	1 movie ~ 5000 s
electric current	ampere	A	USB charging ~2 A. ~0.03 A painful shock.
thermodynamic temperature	kelvin	K	Singapore Room temp: ~300 K
amount of substance	mole	mol	1 mol of carbon ~ 12 g
luminous intensity	candela	cd	Common candle flame ~ 1cd

More information can be found on

<http://physics.nist.gov/cuu/Units/units.html> http://en.wikipedia.org/wiki/SI_base_unit

Other quantities, called *derived quantities*, are defined in terms of the seven base quantities via a system of quantity equations. The *SI derived units* for these derived quantities are obtained from these equations and the seven SI base units.

For example,

area	square meter	m ²
volume	cubic meter	m ³
speed, velocity	meter per second	m/s
acceleration	meter per second squared	m/s ²
wave number	reciprocal meter	m ⁻¹
mass density	kilogram per cubic meter	kg/m ³
current density	ampere per square meter	A/m ²
magnetic field strength	ampere per meter	A/m
amount-of-substance concentration	mole per cubic meter	mol/m ³

Units and Measurements (Pre-lecture Video)

For ease of understanding and convenience, 22 SI derived units have been given special names and symbols (taken from <http://physics.nist.gov/cuu/Units/units.html>): (units that will be used in this course are in **bold**)

Derived quantity	Name	Symbol	Expression in terms of other SI units	Expression in terms of SI base units
plane angle	radian ^(a)	rad	-	$\text{m} \cdot \text{m}^{-1} = 1$ ^(b)
solid angle	steradian ^(a)	sr ^(c)	-	$\text{m}^2 \cdot \text{m}^{-2} = 1$ ^(b)
frequency	hertz	Hz	-	s^{-1}
force	newton	N	-	$\text{m} \cdot \text{kg} \cdot \text{s}^{-2}$
pressure, stress	pascal	Pa	N/m^2	$\text{m}^{-1} \cdot \text{kg} \cdot \text{s}^{-2}$
energy, work, quantity of heat	joule	J	$\text{N} \cdot \text{m}$	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-2}$
power, radiant flux	watt	W	J/s	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-3}$
electric charge, quantity of electricity	coulomb	C	-	$\text{s} \cdot \text{A}$
electric potential difference, electromotive force	volt	V	W/A	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-3} \cdot \text{A}^{-1}$
capacitance	farad	F	C/V	$\text{m}^{-2} \cdot \text{kg}^{-1} \cdot \text{s}^4 \cdot \text{A}^2$
electric resistance	ohm	Ω	V/A	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-3} \cdot \text{A}^{-2}$
electric conductance	siemens	S	A/V	$\text{m}^{-2} \cdot \text{kg}^{-1} \cdot \text{s}^3 \cdot \text{A}^2$
magnetic flux	weber	Wb	$\text{V} \cdot \text{s}$	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-2} \cdot \text{A}^{-1}$
magnetic flux density	tesla	T	Wb/m^2	$\text{kg} \cdot \text{s}^{-2} \cdot \text{A}^{-1}$
inductance	henry	H	Wb/A	$\text{m}^2 \cdot \text{kg} \cdot \text{s}^{-2} \cdot \text{A}^{-2}$
Celsius temperature	degree Celsius	$^{\circ}\text{C}$	-	K
luminous flux	lumen	lm	$\text{cd} \cdot \text{sr}$ ^(c)	$\text{m}^2 \cdot \text{m}^{-2} \cdot \text{cd} = \text{cd}$
illuminance	lux	lx	lm/m^2	$\text{m}^2 \cdot \text{m}^{-4} \cdot \text{cd} = \text{m}^{-2} \cdot \text{cd}$
activity (of a radionuclide)	becquerel	Bq	-	s^{-1}
absorbed dose, specific energy (imparted), kerma	gray	Gy	J/kg	$\text{m}^2 \cdot \text{s}^{-2}$
dose equivalent ^(d)	sievert	Sv	J/kg	$\text{m}^2 \cdot \text{s}^{-2}$
catalytic activity	katal	kat		$\text{s}^{-1} \cdot \text{mol}$

^(a) The radian and steradian may be used advantageously in expressions for derived units to distinguish between quantities of a different nature but of the same dimension; some examples are given in Table 4.

^(b) In practice, the symbols rad and sr are used where appropriate, but the derived unit "1" is generally omitted.

^(c) In photometry, the unit name steradian and the unit symbol sr are usually retained in expressions for derived units.

^(d) Other quantities expressed in sieverts are ambient dose equivalent, directional dose equivalent, personal dose equivalent, and organ equivalent dose.

Vector and Scalar Quantities (Pre-lecture Video)

The physical quantity is either a **vector** or a **scalar**. (Next time you may also learn about tensors but it is beyond the scope of this course.)

Scalars are quantities that are fully described by a magnitude (or numerical value) alone. Vectors are quantities that are fully described by both a magnitude and a direction.

Examples of scalar quantities:

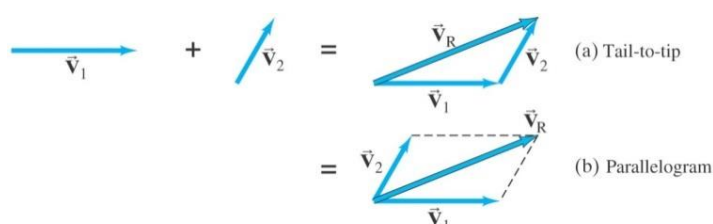
Mass, time, volume, temperature, speed, electric potential.

For scalar quantities, you can add, subtract and do basic arithmetic numerically just as you do with numbers.

Example of vector quantities:

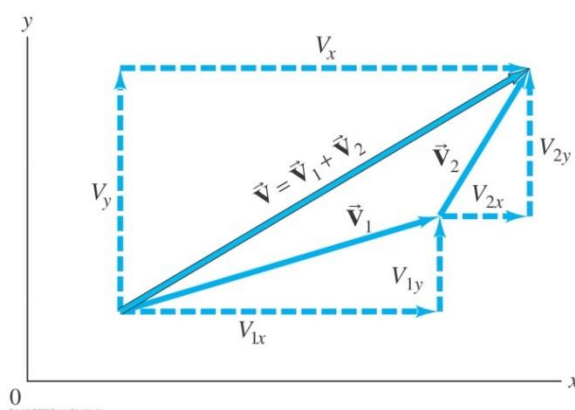
Velocity, acceleration, force, torque, electric field, magnetic field.

For vector quantities, you can only combine the vectors as shown below



If a coordinate system is provided,

Giancoli pg 54 Fig 3-6: Vector Addition



Giancoli pg 56 Fig 3-12: Vector addition in a coordinate system

$$\vec{V} = \vec{V}_1 + \vec{V}_2;$$

$$V_x = V_{1x} + V_{2x};$$

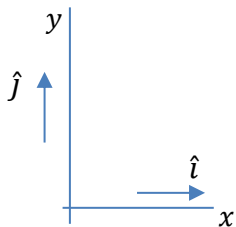
$$V_y = V_{1y} + V_{2y}$$

There are also two types of vector products:

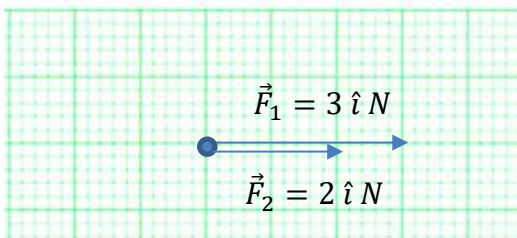
- dot (or also known as scalar) product
- cross (or also known as vector) product

We will discuss these in greater details in the subsequent lectures.

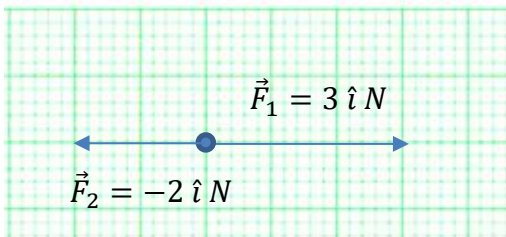
Simple Calculations Involving Vectors (Pre-lecture Video)



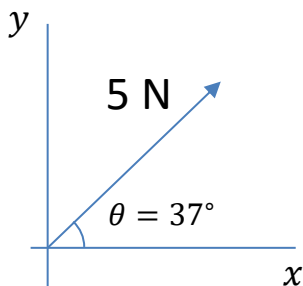
One unit in x direction is $1 \hat{i}$.
One unit in y direction is $1 \hat{j}$.



What is the net force acting on the object?
Write your answer in vector form.

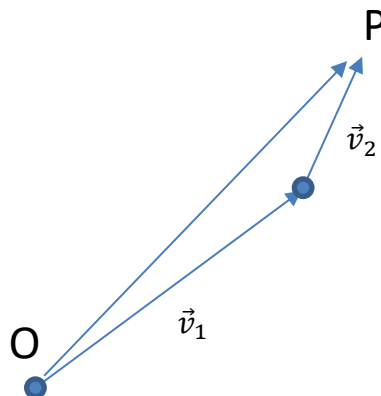


What is the net force acting on the object?
Write your answer in vector form.

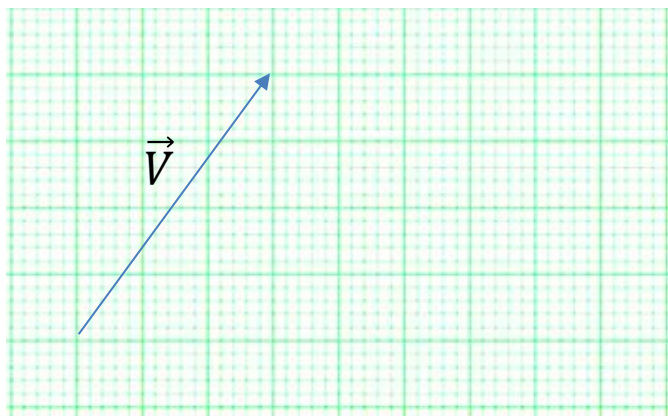


Express the 5N force in vector form
(in $\hat{i}$, $\hat{j}$ components):

$\vec{F} =$



A person starting walking from point O and the position change is given vector $\vec{v}_1$ and then followed $\vec{v}_2$. What is the resultant position change vector $\vec{v}$ (O to P)?



Reading off the grid, write vector $\vec{V}$ in $\hat{i}$, $\hat{j}$ components.

Class Discussion: Video

Basic Vectors

Units and Measurements (Pre-lecture Video)

The 20 SI prefixes used to form decimal multiples and submultiples of SI units are given below:

Factor	Name	Symbol
10^{24}	yotta	Y
10^{21}	zetta	Z
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10^1	deka	da

Factor	Name	Symbol
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a
10^{-21}	zepto	z
10^{-24}	yocto	y



www.scaleofuniverse.com

Exercise:

1. Convert 678978 g to kg; 0.000343 m to mm.

2. Convert 28 m/s to km/hr.

3. Convert 178 cm^2 to m^2 .

4. Convert 2.48 g/cm^3 to kg/m^3 .

Greek Alphabet (Pre-lecture Video)			
Lower case	Capital	Name	Common use in Physics
α	A	alpha	Angular acceleration; Linear expansion Coefficient; Alpha particle (helium nucleus); Fine Structure Constant
β	B	beta	Beta particle — high energy electron Sound intensity
γ	Γ	gamma	Gamma ray (high energy EM wave); Ratio of heat capacities (in an ideal gas); Relativistic correction factor
δ	Δ	delta	Δ="Change in" δ="Infinitesimal change in"
ϵ	E	epsilon	Emissivity; Strain; Permittivity; EMF
ζ	Z	zeta	(no common use)
η	H	eta	Viscosity; Energy efficiency
θ	Θ	theta	Angle ($^\circ$, rad); Temperature
ι	I	iota	The lower case ι is rarely used, while I is sometimes used for the identity matrix or the moment of inertia. Note that ι is not to be confused with the Roman character i ; i (which has a dot and is much more widely used in mathematics and physics).
κ	K	kappa	Spring constant; Dielectric constant
λ	Λ	lambda	Wavelength; Thermal conductivity Constant Eigenvalue of a matrix; Linear density
μ	M	mu	Coefficient of friction; Electrical mobility; Reduced mass; Permeability
ν	N	nu	Frequency
ξ	Ξ	xi	Damping coefficient
$\omicron$	O	omicron	-
π	Π	pi	Product symbol Π Circle number $\pi := 3.14159...$
ρ	ρ	rho	Volume density; Resistivity
σ	Σ	sigma	Sum symbol ; Boltzmann constant Electrical conductivity; Uncertainty; Stress; Surface density
τ	T	tau	Torque; Time constant; Tau particle (a lepton);
υ	Y	upsilon	mass to light ratio
ϕ	Φ	phi	Magnetic/electric flux; Angle ($^\circ$, rad)
χ	X	chi	Rabi frequency (lasers); Susceptibility
ψ	Ψ	psi	Wave function
ω	Ω	omega	Ohms (unit of electrical resistance) ω Angular velocity

Source: https://en.wikibooks.org/wiki/Physics_Study_Guide/Greek_alphabet

Homogeneity of Units (Pre-lecture Video)

Given an equation

$$A + B + \frac{C}{D} + E \times F = G$$

The units of $A, B, \frac{C}{D}, E \times F$ and G must be the same.

Mathematical functions can only operate on pure numbers with no physical units. For example, $\cos \theta$, here θ has no units (rad is not a physical unit). Other examples include $\ln x$ and e^x . Here, x must be a number with not units.

Example

Newton's law of gravitation states that the mutual force of attraction between two objects of masses M_1 and M_2 separated by a distance r is given by

$$F = \frac{GM_1M_2}{r^2}$$

where G is the universal gravitational constant. Deduce the SI unit of G .

Example

An equation often used in fluid mechanics, known as Bernoulli's equation is given by

$$\frac{1}{2}\rho v^2 + \rho gz + P = \text{constant}$$

Here, v is velocity in m/s. The unit for density ρ is in kg/m³ and the unit for acceleration of free fall g in m/s². What are the units for the variables z and P ?

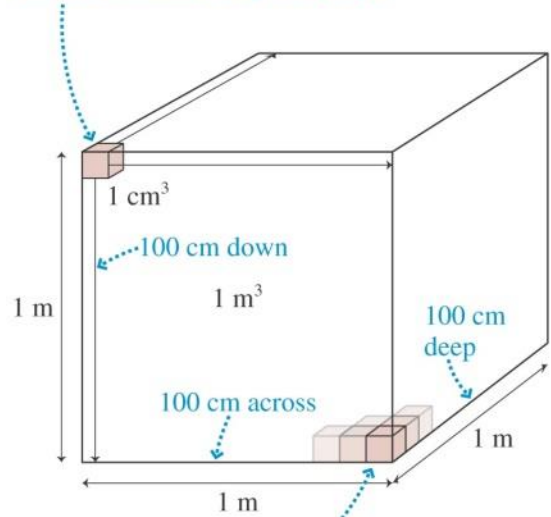
Visualizing Physical Quantities and Units (Pre-lecture Video)



1 litre – about 1 kg

Volume of 1 m^3

Subdivide the $1 \text{ m} \times 1 \text{ m} \times 1 \text{ m}$ cube into little cubes 1 cm on a side. You will get 100 subdivisions along each edge.



There are $100 \times 100 \times 100 = 10^6$ little 1 cm^3 cubes in the big 1 m^3 cube.

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Knight Fig 15.2

Weight of 1 apple $\sim 1 \text{ N}$

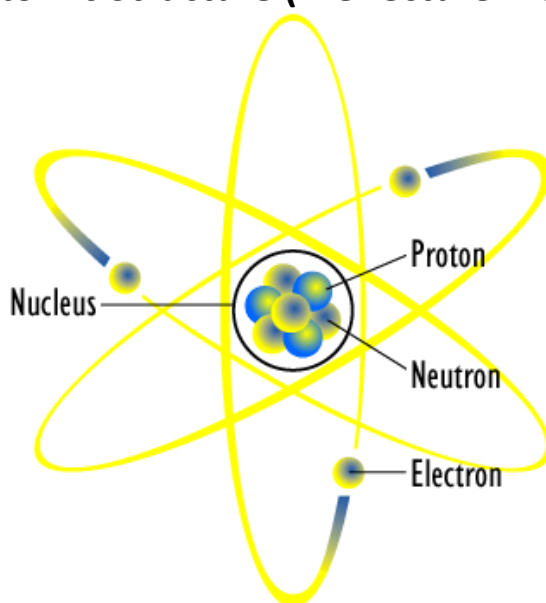


100 m in about 10 s $\Rightarrow \sim 10 \text{ m/s} \equiv 36 \text{ km/h}$



<https://www.gettyimages.com/detail/news-photo/usain-bolt-of-jamaica-competes-in-the-mens-100-meter-news-photo/589521390?adppopup=true>

Atomic Structure (Pre-lecture Video)



http://en.wikipedia.org/wiki/Rutherford_model

Matter is made up of atoms. Each atom is made of *protons* and *neutrons* which forms its core – the *nucleus* with electrons moving round it.

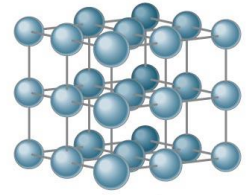
	Mass (kg)	Charge (C)
Proton	1.6726×10^{-27}	$+1.602 \times 10^{-19}$
Neutron	1.6749×10^{-27}	0
Electron	9.109×10^{-31}	-1.602×10^{-19}

In an electrically neutral atom, the number of electrons is the same as the number of protons. The size of the nucleus is about 10^{-15} m and the size of the atom is about 10^{-10} m.

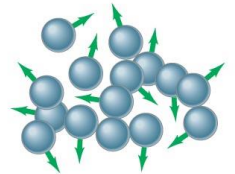
Atomic Structure of Solid, Liquid and Gases (Pre-lecture Video)

On a microscopic scale, the arrangements of molecules in solids (a), liquids (b), and gases (c) are quite different.

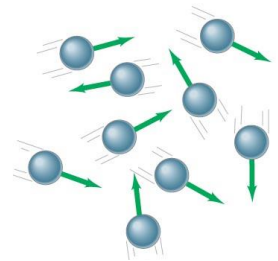
Particles in a solid are tightly packed, usually in a regular pattern. They vibrate (jiggle) but generally do not move from place to place.



Particles in liquid state are close together with no regular arrangement. They liquid vibrate, move about, and slide past each other.



Particles in a gaseous state are well separated with no regular arrangement. They vibrate and move freely at high speeds.



Liquids and solids are often referred to as **condensed phases** because the particles are very close together.

<http://www.chem.purdue.edu/gchelp/liquids/character.html>

Brownian Motion

Giancoli Fig 17.2

http://en.wikipedia.org/wiki/Brownian_motion

Thermal Expansion (from perspective of a fixed central atom)

<http://web.mit.edu/mbuehler/www/SIMS/Thermal%20Expansion.html>

Mass and Weight

Mass is the measure of inertia of an object, sometimes understood as the quantity of matter in the object. In the SI system, mass is measured in kilograms.

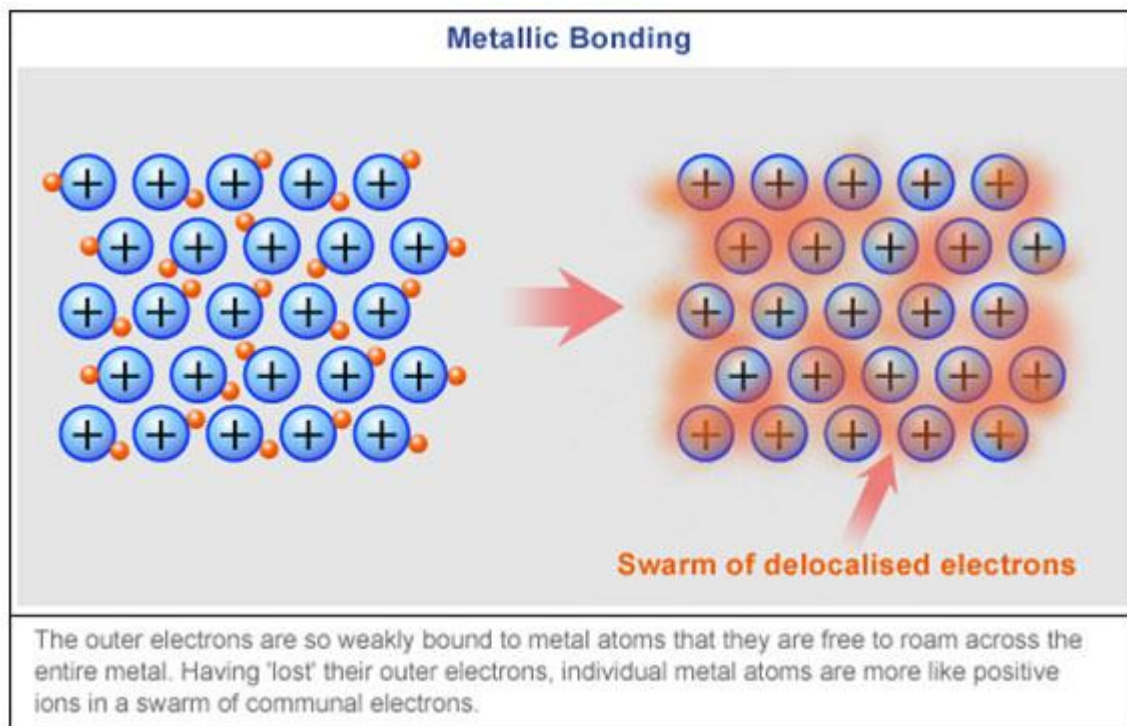
Mass is not weight.

Mass is a property of an object. Weight is the force exerted on that object by gravity. The gravitational field strength near the surface of Earth $g = 9.81 \text{ N/kg}$.

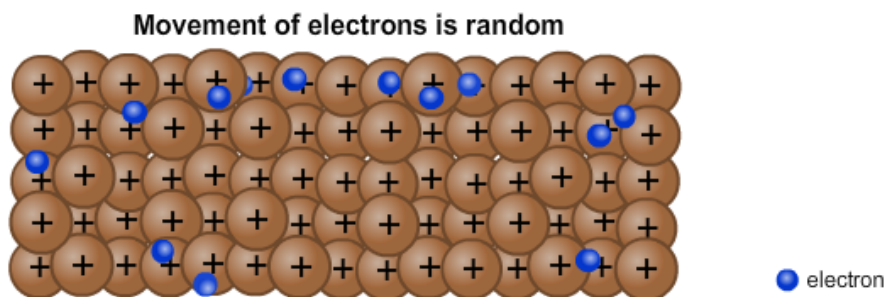
For example, if the mass of an object is 5 kg, then its weight is $5 \times 9.81 = 49.05 \text{ N}$. Weight is a force and a vector. It has both magnitude and direction.

If you go to the Moon, whose gravitational acceleration is about $1/6 g$, you will weigh much less. Your mass, however, will be the same.

Metals (Pre-lecture Video)



<http://www.texasgateway.org/resource/chemical-bonding-metallic-bonds>



- Atoms in a metal are arranged in a structured manner and vibrate in place.
- The outermost electrons (valence electrons) move freely, forming a "sea of electrons" that surrounds the positive metal ions.
- The metal cations are attracted to the sea of electrons. This attraction holds the ions together. This makes up the metallic bond.
- The movements of electrons are random under normal conditions.

Density

The density ρ of a substance is its mass per unit volume:

$$\rho = \frac{m}{V}$$

The SI unit for density is kg/m^3 . Density is also sometimes given in g/cm^3 ; to convert g/cm^3 to kg/m^3 , multiply by 1000.

Water at 4°C has a density of $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$.

The specific gravity of a substance is the ratio of its density to that of water.

Giancoli Prob 15.3

The dimensions of a piece of gold is $56 \text{ cm} \times 28 \text{ cm} \times 22 \text{ cm}$. What would the mass be?

Giancoli Prob 13.5

A bottle has a mass of 35.00 g when empty and 98.44 g when filled with water. When filled with another fluid, the mass is 89.22 g. What is the specific gravity of this other fluid?



35.00 g



98.44 g



89.22 g

$$\text{Spec grav} = \frac{\rho_x}{\rho_w} =$$

Giancoli Table 13.1

TABLE 13-1
Densities of Substances[†]

Substance	Density, ρ (kg/m^3)
<i>Solids</i>	
Aluminum	2.70×10^3
Iron and steel	7.8×10^3
Copper	8.9×10^3
Lead	11.3×10^3
Gold	19.3×10^3
Concrete	2.3×10^3
Granite	2.7×10^3
Wood (typical)	$0.3 - 0.9 \times 10^3$
Glass, common	$2.4 - 2.8 \times 10^3$
Ice (H_2O)	0.917×10^3
Bone	$1.7 - 2.0 \times 10^3$
<i>Liquids</i>	
Water (4°C)	1.00×10^3
Blood, plasma	1.03×10^3
Blood, whole	1.05×10^3
Sea water	1.025×10^3
Mercury	13.6×10^3
Alcohol, ethyl	0.79×10^3
Gasoline	0.68×10^3
<i>Gases</i>	
Air	1.29
Helium	0.179
Carbon dioxide	1.98
Steam (water, 100°C)	0.598

[†]Densities are given at 0°C and 1 atm pressure unless otherwise specified.

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<https://www.sigmaaldrich.com/deepweb/assets/sigmaaldrich/product/images/185/795/ca0f0f06-b1f1-49e9-9a18-cc6bf225fea4/800/ca0f0f06-b1f1-49e9-9a18-cc6bf225fea4.jpg>

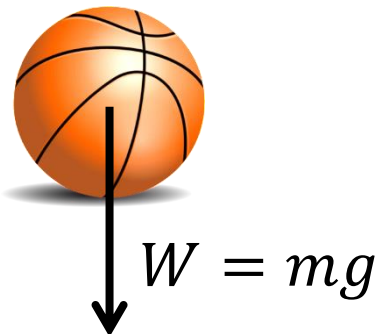
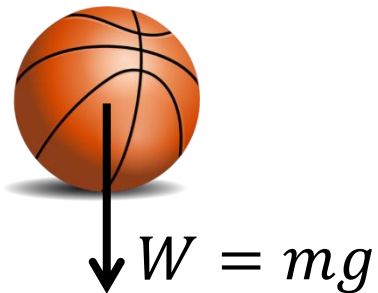
Three Basic Forces

Weight W is the force exerted on an object by gravity. Near the surface of the Earth, where the gravitational field is nearly constant, the weight of an object of mass m is:

$$W = mg$$

where

$$g = 9.8 \text{ m/s}^2$$



This force acting on the ball at the point of contact is known as the '**Normal Force**' F_N . It is a force due to contact and acts perpendicular to the surface.

When the ball is in equilibrium, an equal and opposite normal force F_N is acting downwards on the ground.

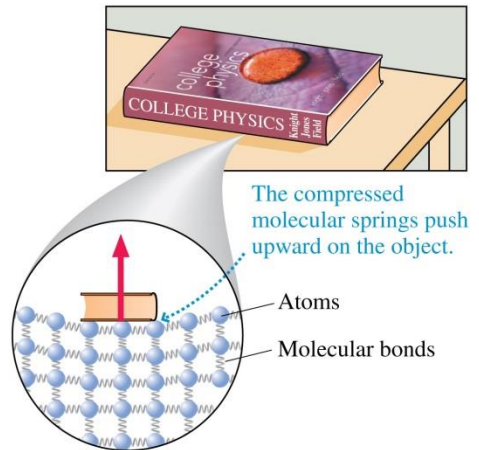


Force is a vector. It has a magnitude and direction.

Normal Force

- The force exerted on an object that is pressing against a surface is in a direction *perpendicular* to the surface.
- The **normal force** is the force exerted by a surface (the agent) against an object that is pressing against the surface.

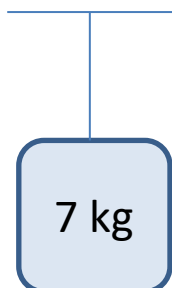
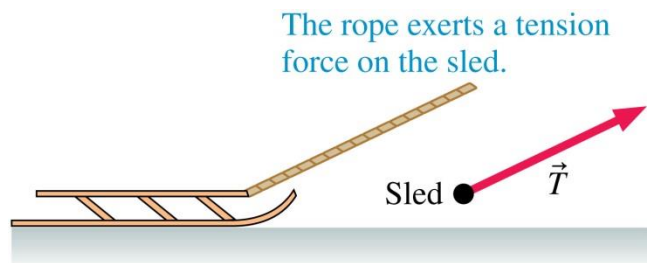
The book in the diagram has a mass of 1.2 kg and is resting on the table top. Calculate the normal force exerted by the table top on the book.



Tension Force

When a string or rope or wire pulls on an object, it exerts a contact force that we call the **tension force**.

- **The direction of the tension force is always in the direction of the string or rope.**
- The direction of the tension is always pointing away from the object.



A 7kg block hanging from the ceiling via a cord. What is the tension in the cord?

Pressure

Pressure is defined as the force per unit area.

$$\text{pressure} = P = \frac{F}{A}.$$

Pressure is a scalar; the units of pressure in the SI system are pascals:

$$1 \text{ Pa} = 1 \text{ N/m}^2.$$

Giancoli Table 13.2

TABLE 13–2 Conversion Factors Between Different Units of Pressure	
In Terms of $1 \text{ Pa} = 1 \text{ N/m}^2$	1 atm in Different Units
$1 \text{ atm} = 1.013 \times 10^5 \text{ N/m}^2$ $= 1.013 \times 10^5 \text{ Pa} = 101.3 \text{ kPa}$	$1 \text{ atm} = 1.013 \times 10^5 \text{ N/m}^2$
$1 \text{ bar} = 1.000 \times 10^5 \text{ N/m}^2$	$1 \text{ atm} = 1.013 \text{ bar}$
$1 \text{ dyne/cm}^2 = 0.1 \text{ N/m}^2$	$1 \text{ atm} = 1.013 \times 10^6 \text{ dyne/cm}^2$
$1 \text{ lb/in.}^2 = 6.90 \times 10^3 \text{ N/m}^2$	$1 \text{ atm} = 14.7 \text{ lb/in.}^2$
$1 \text{ lb/ft}^2 = 47.9 \text{ N/m}^2$	$1 \text{ atm} = 2.12 \times 10^3 \text{ lb/ft}^2$
$1 \text{ cm-Hg} = 1.33 \times 10^3 \text{ N/m}^2$	$1 \text{ atm} = 76.0 \text{ cm-Hg}$
$1 \text{ mm-Hg} = 133 \text{ N/m}^2$	$1 \text{ atm} = 760 \text{ mm-Hg}$
$1 \text{ torr} = 133 \text{ N/m}^2$	$1 \text{ atm} = 760 \text{ torr}$
$1 \text{ mm-H}_2\text{O} (4^\circ\text{C}) = 9.80 \text{ N/m}^2$	$1 \text{ atm} = 1.03 \times 10^4 \text{ mm-H}_2\text{O} (4^\circ\text{C})$

Copyright © 2008 Pearson Education, Inc.

Question: The dimensions of a piece of gold is $56 \text{ cm} \times 28 \text{ cm} \times 22 \text{ cm}$. It is placed flat on a table top. What is the lowest pressure that it can exert on the table top?

Pressure in Fluids

Giancoli Chap 13

For a fluid at rest, the force due to the fluid pressure always act perpendicular to any solid surfaces it touches.

Pressure is the same in every direction in a static fluid at a given depth; if it were not, the fluid would flow.

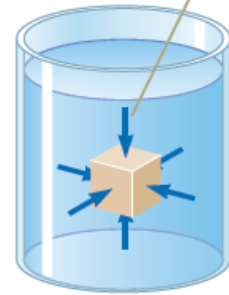
For a fluid at rest, there is also no component of force parallel to any solid surface—once again, if there were, the fluid would flow.

The pressure at a depth h below the surface of the liquid is due to the weight of the liquid above it. We can quickly calculate:

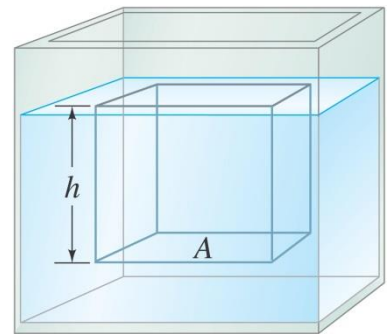
$$P = \frac{F}{A} = \frac{\rho Ahg}{A}$$

$$P = \rho gh.$$

At any point on the surface of the object, the force exerted by the fluid is perpendicular to the surface of the object.



Serway Fig 14.1: The forces exerted by a fluid on the surfaces of a submerged object.



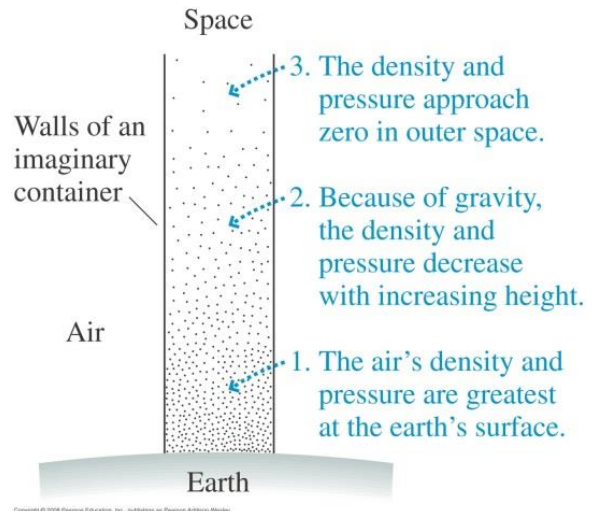
Atmospheric Pressure

This relation is valid for any liquid whose density does not change with depth.

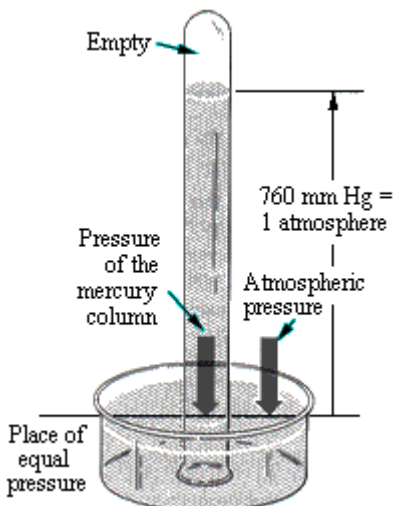
At sea level the atmospheric pressure is about $1.013 \times 10^5 \text{ N/m}^2$; this is called 1 atmosphere (atm). Another unit of pressure is the bar:

$$1 \text{ bar} = 1.00 \times 10^5 \text{ N/m}^2.$$

Standard atmospheric pressure is just over 1 bar. This pressure does not crush us, as our cells maintain an internal pressure that balances it.



Knight Fig 15.9



At sea level the atmospheric pressure can balance the pressure due to a column of mercury of height 760 mm.

<http://www.chem.uiuc.edu/rogers/Text9/Tx94/tx94.html>

Most pressure gauges measure the pressure above the *atmospheric pressure*—this is called the *gauge pressure*. The *absolute pressure* is the sum of the *atmospheric pressure* and the *gauge pressure*.

$$P = P_0 + P_G.$$



https://commons.wikimedia.org/wiki/File:Canadian_Pacific_pressure_gauge_%282800301396%29.jpg

Class Discussion

Some divers can dive to a depth of 20m without scuba diving tanks. What is the pressure experienced by a diver who is diving at 20 m under seawater?

(Here, we assume that the density of seawater to be 1000 kg/m^3 and $g = 10 \text{ m/s}^2$ for simplicity.)

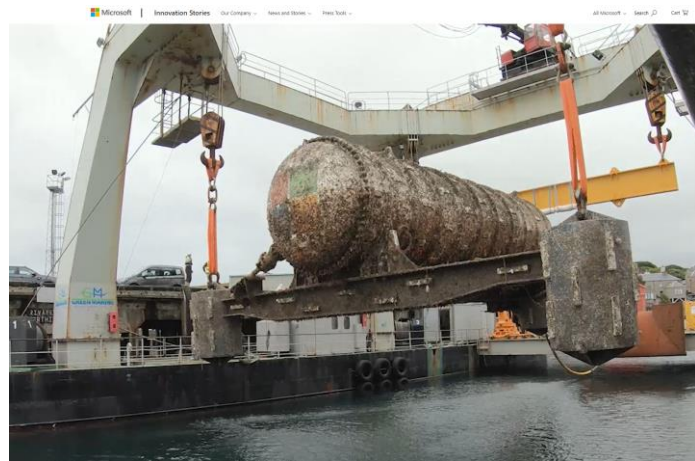


How many times is this compared to the atmospheric pressure?

<https://www.youtube.com/watch?v=xJHJsA7bYGc&t=5s>

Example: Microsoft finds underwater data centers are reliable, practical and use energy sustainably

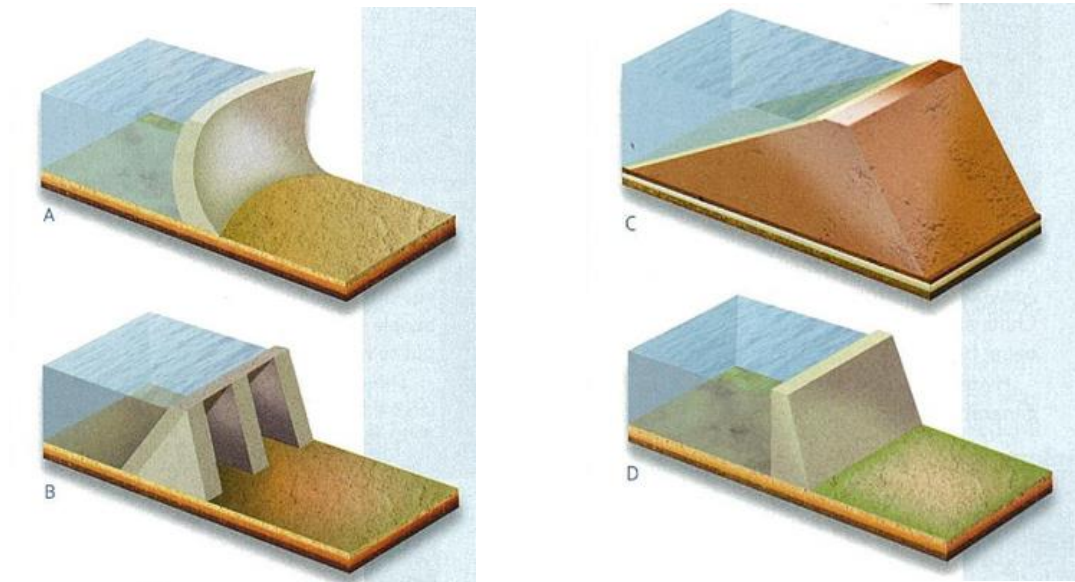
“Earlier this summer, marine specialists reeled up a shipping-container-size datacenter coated in algae, barnacles and sea anemones from the seafloor off Scotland’s Orkney Islands. The retrieval launched the final phase of a years-long effort that proved the concept of underwater datacenters is feasible, as well as logistically, environmentally and economically practical. Microsoft’s [Project Natick](#) team [deployed the Northern Isles datacenter](#) 117 feet deep to the seafloor in spring 2018. For the next two years, team members tested and monitored the performance and reliability of the datacenter’s servers....”



Source: <https://news.microsoft.com/innovation-stories/project-natick-underwater-datacenter/>

Determine the absolute pressure at a depth of 117 feet.

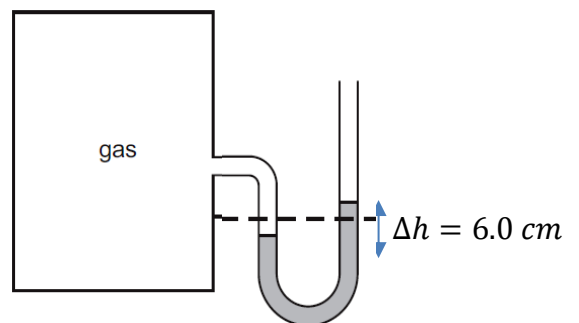
Dams



<http://www.australiangeographic.com.au/topics/science-environment/2011/01/to-dam-or-not-to-dam/>

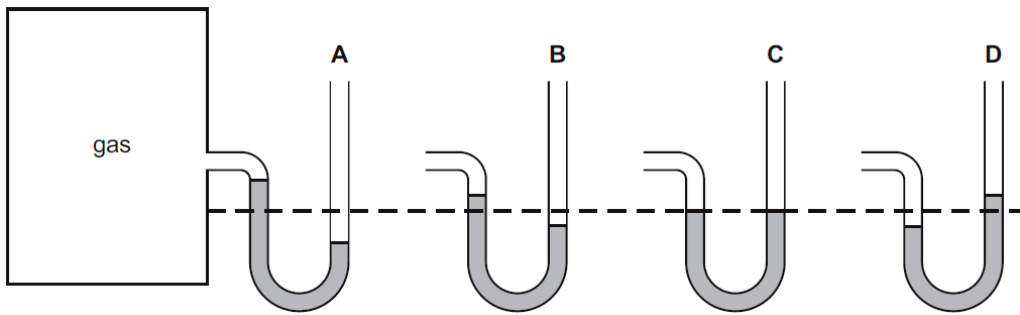
Exercise:

The mercury in the tube is in equilibrium and the height difference between the mercury level in the two limbs is 6.0 cm. Determine the gas pressure in the tank. Density of mercury is $13\,600\text{ kg m}^{-3}$.



Class Discussion:

In which of the following is the gas pressure the highest?



Upthrust

Singapore now home to one of the world's largest floating solar farms



The 5MW-peak system installation is expected to produce an estimated 6 million kilowatt hours of energy per year. PHOTO: SUNSEAP GROUP



Clara Chong

UPDATED: MAR 24, 2021, 7:07 AM



SINGAPORE - Singapore is now home to one of the world's largest floating solar farms and it is capable of potentially offsetting more than 4,000 tonnes of carbon dioxide per year.

<https://www.straitstimes.com/singapore/singapore-now-home-to-one-of-the-worlds-largest-floating-solar-farms>

An aerial view of a floating data centre structure in the water. The structure consists of several rectangular platforms connected by a network of walkways and bridges. The water is dark blue, and the sky is light blue. A play button icon is overlaid on the image, indicating it is a video thumbnail.

Singapore

Floating data centre being developed to overcome land, energy constraints | Video

31 Aug 2019 10:59PM
(Updated: 07 Nov 2020 04:18PM)

Floating data centres at sea may be the way forward to overcome the land, water and energy constraints of such facilities. Keppel Data Centres said it has a concept which it is ready to deploy.

Bookmark

Dell National Day Sale

Boost Your Business with Dell National Day Sale, and Up to \$350 cash off* (selected Vostro PCs, T6Cs apply)

SEE MORE

<https://www.channelnewsasia.com/news/singapore/floating-data-centre-being-developed-to-overcome-land-energy-11860550>

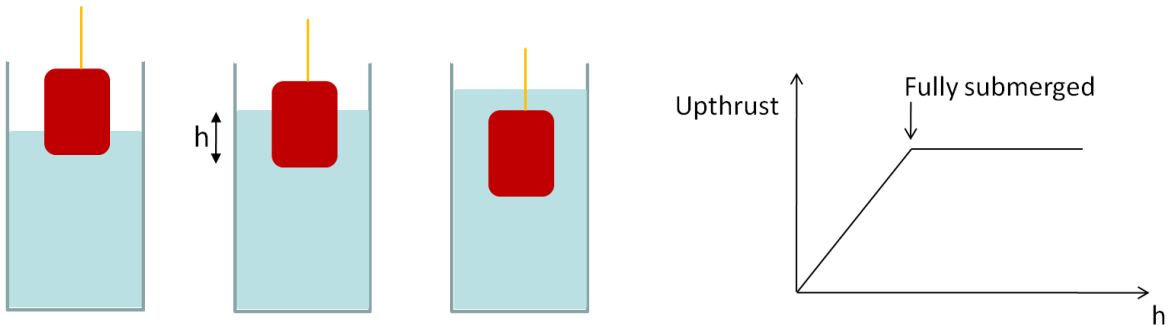
Upthrust

The density of iron is 7.86 g cm^{-3} . The density of sea water to be 1.10 g cm^{-3} . Can iron float in sea water?



<https://www.youtube.com/watch?v=05WkCPORlj4&t=1s>

When an object is partially immersed in a fluid, it will experience an upward force. When the object is immersed further in a fluid, the upward force will increase correspondingly until it is fully immersed. This force is known as upthrust. It originates from the pressure difference between the bottom (larger pressure) and top side of the object.



Archimedes' Principle

Any object, wholly or partially immersed in a fluid, is buoyed up by a force equal to the weight of the fluid displaced by the object.

Remarks:

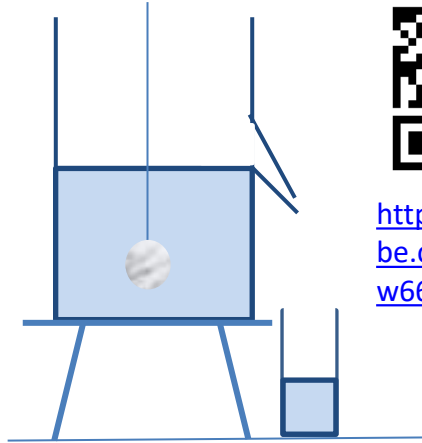
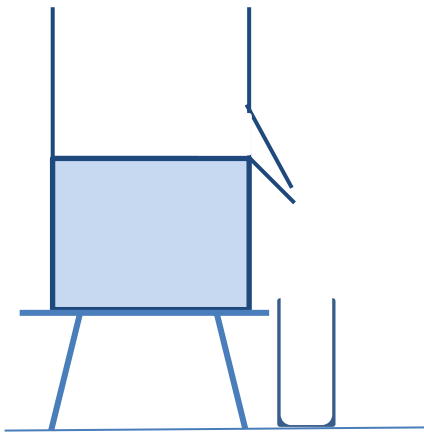
1. Volume of object submerged in the fluid = Volume of fluid displaced;
2. Knowing the displaced volume $V_{\text{submerged}}$ and density ρ_{fluid} of the fluid, we can compute the weight of fluid which gives the upthrust:

$$\text{Upthrust} = (\rho_{\text{fluid}} V_{\text{submerged}}) g$$

From here, we can derive the law of flotation: A floating object displaces its own weight of fluid. For example, if an object weighs 1.2 N, for it to float, it has to be able to displace at least 1.2 N of fluid, i.e. giving an upthrust equivalent to 1.2 N.

HOSY@ntu.edu.sg

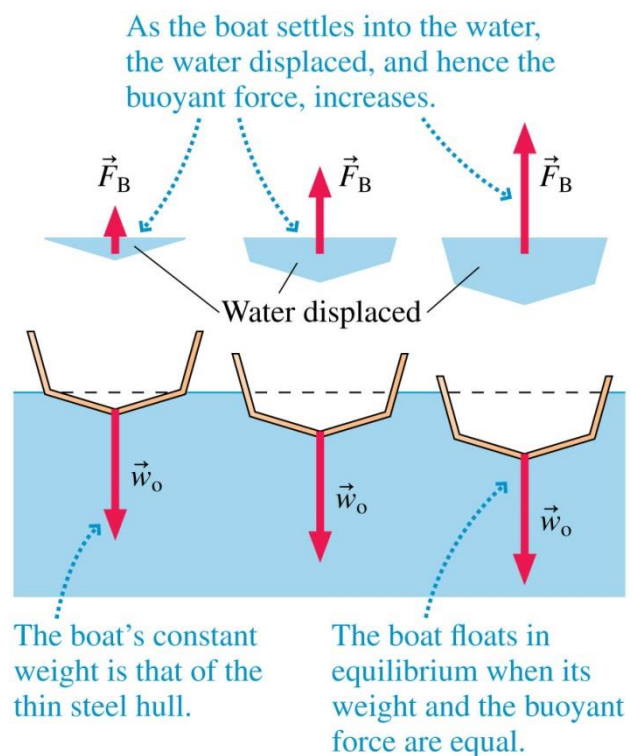
Displacement Can



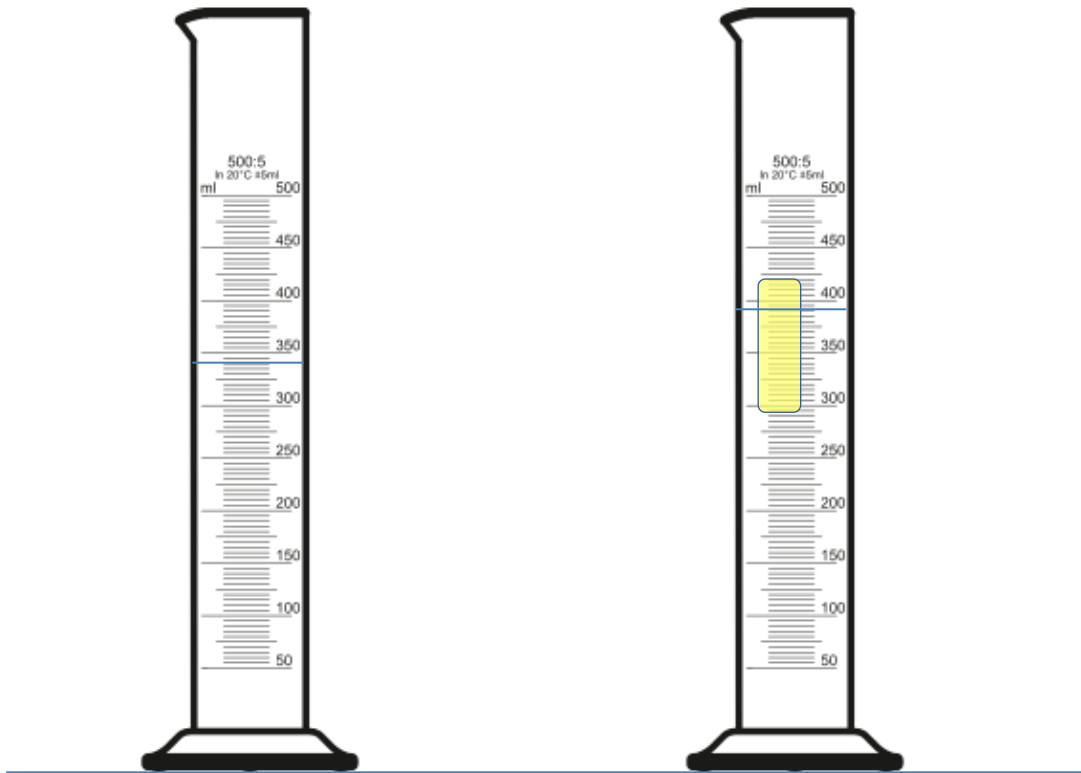
<https://www.youtube.com/watch?v=sW66gzBD9fE&t=4s>

Boats (taken from Knight)

The hull of a boat is really a hollow shell, so the volume of water displaced by the shell is *much* larger than the volume of the hull itself. The boat sinks until the weight of the displaced water exactly matches the boat's weight. It is then in static equilibrium and floats.



Test your understanding



A piece of material with density 0.70 g cm^{-3} is partially immersed in oil with density 0.85 g cm^{-3} .

- i. What is the volume of oil displaced by the block of material?
- ii. What is the weight of oil displaced by the block of material?
- iii. What is the upthrust acting on the block due to oil?
- iv. If someone use his finger to push the block further down into the liquid, how would answer (i) to (iii) change?

Example



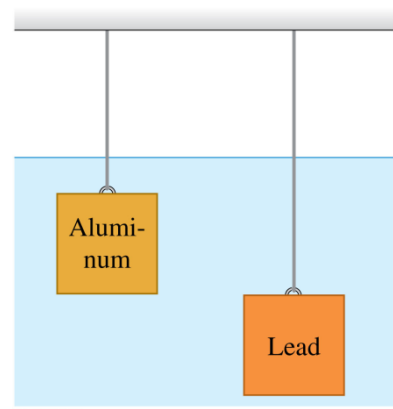
If the duck displaces 0.25 m^3 of water (assuming density $\rho = 1.00 \text{ g/cm}^3$), calculate the upthrust acting on it.

http://en.wikipedia.org/wiki/Rubber_duck#media:File:Rubber_Duck_%288374802487%29.jpg

Question (Young and Freedman Q12.30)

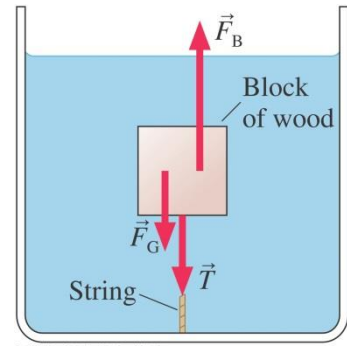
Identical sized lead and aluminum cubes are suspended at different depths in a large vat of water (density of lead is higher).

- a) Which cube experiences a greater buoyant force?
- b) For which cube is the tension in the string greater?



Knight (2) Example 15.8

A $10\text{ cm} \times 10\text{ cm} \times 10\text{ cm}$ block of wood with density 700 kg/m^3 is held underwater by a string tied to the bottom of the container. What is the tension in the string? Density of water is 1000 kg/m^3 .



Class Discussion: Example: 2020 Mid-term 1 Q

The figure on the right shows an aerostat (filled with helium) and can lift antennas and receivers to an altitude of 250 meters and deliver mobile broadband to rural communities. The aerostat is tethered to the ground by a system of cables. One such system can provide ground coverage for up to $10,000\text{ km}^2$, an area that would normally require about 15 cell phone towers – reducing the deployment costs by 60%. It is a viable candidate to deliver the 5G networks.



The density of helium at 1.0 atm and 15°C is 0.168 kg/m^3 and the density for air is 1.225 kg/m^3 . For a particular system, the mass of the balloon (skin without helium), cables and the equipment is 650 kg . The volume of the balloon is 800 m^3 when in operation. Determine the tension in the cables required to hold the aerostat at a constant height.

2026S1 PH1012: Physics A
Basic Thermal Physics

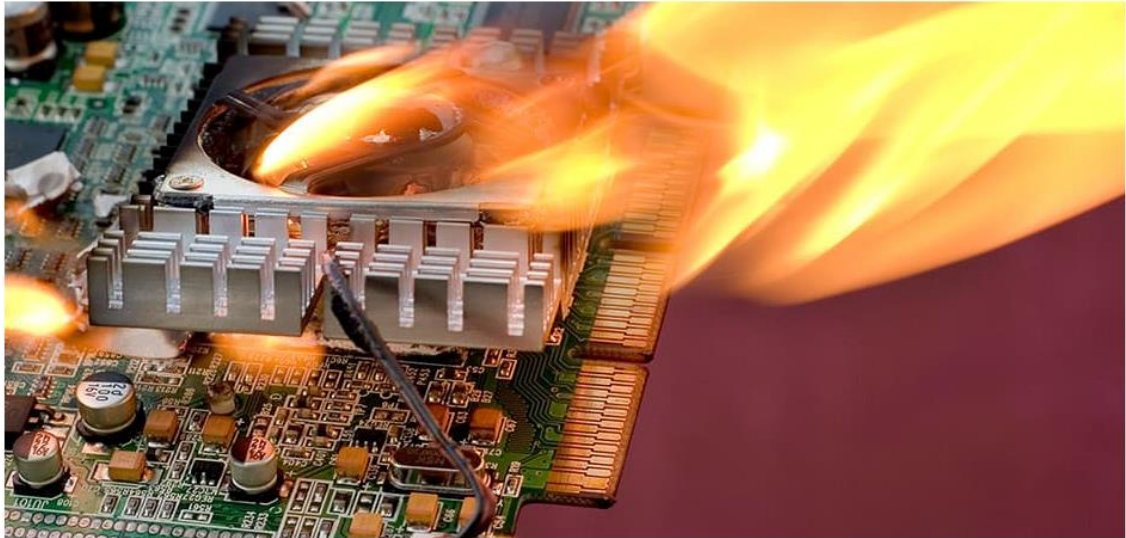
Dr Ho Shen Yong
Lecturer, School of Physical and Mathematical Sciences
Nanyang Technological University

Weeks 1 & 2
Swerway Chap 18, 19.1 – 19.3

"Enthusiasm spells the difference between mediocrity and accomplishment."
- Norman Vincent Peale (1898 – 1993)

Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. calibrate thermometers according to thermometric properties (linear)
2. apply the ideal gas equation to solve problems
3. solve problems related to thermal expansion (length, area, volume)
4. apply the concept specific heat capacity / heat capacity to solve problems
5. apply concepts of energy conservation and heat capacity to solve problems.
6. synthesize the concepts learnt to solve problems.



 Home



How to Check Your Computer's CPU Temperature

Dwight Pavlovic | July 22, 2020

If you've used a PC, chances are you've experienced a spike in central processing unit (CPU) temperature at some point. Sometimes, it's the result of aging hardware. But it can also happen during periods of heavy use.

<https://www.hp.com/gb-en/shop/tech-takes/how-to-check-cpu-temperature>

ChannelnewsAsia: Greening the Net

<https://www.channelnewsasia.com/news/video-on-demand/greening-the-net/greening-the-net-13994106>

About the show:

Life has gotten more digital than ever before. And the plusses of that for the environment has been clear...as in clear skies and clean air. What is not so clear however is this: when it comes to greenhouse gases, the Internet is responsible for 2% of global emissions. If the Internet were a country, it would be ranked as the sixth largest polluter in the world.

But the Internet is here to stay, so what can be done to make it greener? From individuals not hitting Reply All on emails or turning off Auto Play for videos, to data centres running on green energy and search engines giving back to the environment, host Prerna Pant looks at all the various ways we are Greening The 'Net.

Explained: Generative AI's environmental impact

<https://news.mit.edu/2025/explained-generative-ai-environmental-impact-0117>

Does using AI impact the environment? - BBC World Service

<https://www.youtube.com/watch?v=bWeUJxQnyoo>

Temperature (Pre-lecture Video)

Temperature is the measure of the degree of 'hotness' or 'coldness' of a system. However, this description is not objective and we have a more scientific and objective definition. Temperature is a scalar quantity.

Zeroth Law of Thermodynamics:

If systems A and B are each in thermal equilibrium with a third system C , then A and B are in thermal equilibrium with each other.

Remarks:

1. Zeroth law defines the concept of temperature.
2. When two systems are in thermal equilibrium, the net heat flow between the two systems is zero. We can say that they have the same *temperature*.

Temperature and thermometric properties:

Unlike mass, length and time etc, there is no direct way of quantifying temperature as it is a property of collective microscopic behaviour. We have to rely on other measurable physical properties that vary with temperature to determine temperature indirectly.

One such property is the volume of liquids. Other examples include the lengths of metals, resistance of metals, difference of e.m.f.s (thermocouple), pressure of a fixed volume of gas and colour of chemicals.

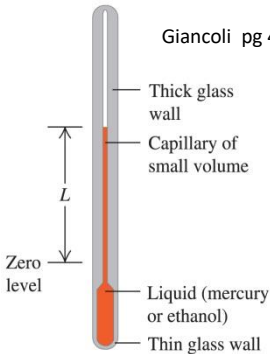
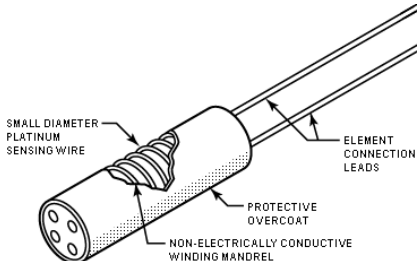
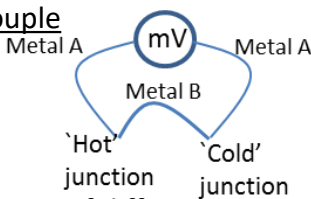
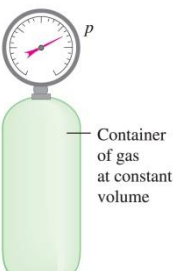
Thus, we have a measurable property P that varies with temperature T :

$$P(T) = A + BT + CT^2 + \dots$$

Or sometimes, the change of the property ΔP varies,

$$\Delta P(T) = A + B\Delta T + \dots$$

Gallery of Thermometers (Pre-lecture Video)

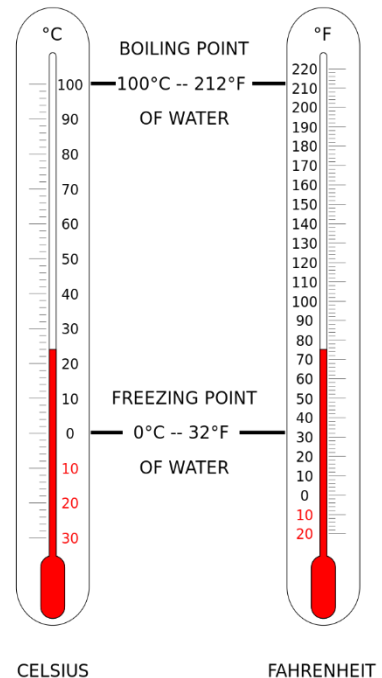
	Properties	Range / Response / Accuracy
 Giancoli pg 422, Fig 14.1	<u>Mercury in glass thermometer</u> Relies on variation of volume of mercury with temperature. Mercury is opaque and easily seen; good conductor of heat; does not stick to glass. $V = V_0 (1 + \beta \Delta T)$	a) -39°C to 357°C (can be increased with modifications). b) Slow response (relatively large heat capacities). c) Typically 0.1°C. (Non-uniform bore, expansion of glass.) Affects temp of object it is measuring.
 http://www.rdfcorp.com/	<u>Platinum Resistance thermometer</u> Rely on the fact that the electrical resistance of metals are temp dependent. Platinum has temp coef of resistance; high melting pt (1773°C). $R(T) = R_0 (1 + \alpha T + \beta T^2)$	a, c) Extremely accurate from -200°C to 1200°C. b) Relatively large heat capacities so takes longer time to come in thermal equilibrium with surroundings – slow response.
 http://www.hcs77.com/Barnant_dualog.htm	<u>Thermocouple</u>  Two fine wires of different metals – e.m.f E in millivoltmeter depends on temp diff at junction $E(T) = aT + bT^2$	Can set cold junction in ice/water (0°C). a) Using several combinations of metals can get from -269°C to 2300°C. b) Small heat capacity – fast response and can measure temp even at embedded pt. c) Accurate over wide range.
 (b) Changes in temperature cause the pressure of the gas to change. Young (College) pg 422, Fig 14.1	<u>Constant Vol Gas thermometer</u> Kept at constant vol, the pressure of the gas varies with temp. For ideal gas, $PV = nRT$. All other thermometers depend critically on the nature and purity of materials used. Under the right conditions, behaviour of gas thermometer is independent of gas.	Seldom used any more as thermometers - used to define meaning of temp. Used as standard reference for temp. a) About -271°C to 1100°C b) Very slow response (large vol of gas used). c) Accurate over a wide range

Calibration of Thermometers; Fixed points (Pre-lecture Video)

In order to establish a temperature scale, it is necessary to make use of fixed points. At each fixed point, a single temperature corresponds to a particular physical phenomena that can be easily and accurately reproduced. At the fixed points, the temperature of all thermometers agree.

Three such fixed points are:

1. Ice point – temp at which pure ice and water co-exist in equilibrium at atmospheric pressure. (0°C ; $\sim 101\text{ kPa}$)
2. Steam point – temp at which pure water and steam co-exist in equilibrium at atmospheric pressure. (100°C ; $\sim 101\text{ kPa}$)
3. Triple point of water – temp at which pure ice, water and water vapour can exist together in equilibrium. (273.16 K , 0.01°C ; 611.73 Pa)



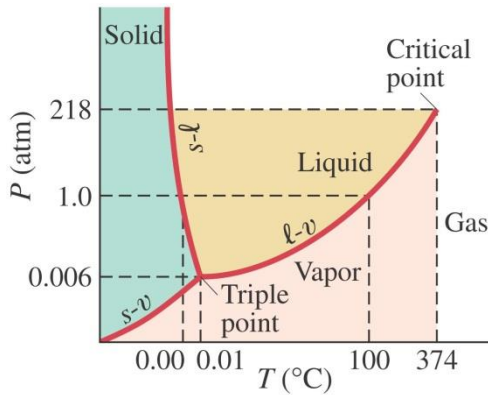
http://en.wikipedia.org/wiki/Thermometer#mediaviewer/File:Thermometer_CF.svg

Giancoli Prob 17.6

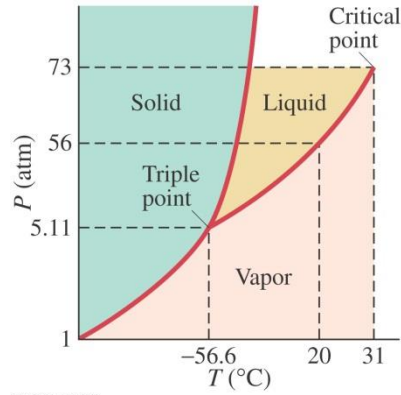
In an alcohol-in-glass thermometer, the alcohol column has length 11.82 cm at 0.0°C and length 21.85 cm at 100.0°C . What is the temperature if the column has length (a) 18.70 cm, and (b) 14.60 cm?

Calibration of Thermometers; Fixed points (Pre-lecture Video)

Triple point of water – temp at which pure ice, water and water vapour can exist together in equilibrium. (273.16 K, 0.01 °C; 611.73 Pa)



Giancoli pg 483 Fig 18.5:
Phase diagram for water



Giancoli pg 483 Fig 18.6:
Phase diagram for CO_2

Remarks:

1. The triple point is special as there is only one pressure at which all three phases of water can be together where ice/water and water/steam can co-exist over a wide range of pressure.
2. The S.I. unit for temperature is *Kelvin* (K).
3. The degree Celsius (°C) is related to the Kelvin scale

$$\theta (^{\circ}C) = T (K) - 273.15$$

a temperature change of 1 K is equal to a temperature change of 1 °C.

Google Homework: What is difference between Centigrade and Celsius scale?)

Youtube homework: Watch “Science! - Cyclohexane at the Triple Point”



<https://www.youtube.com/watch?v=MP6MVLWuNZQ>



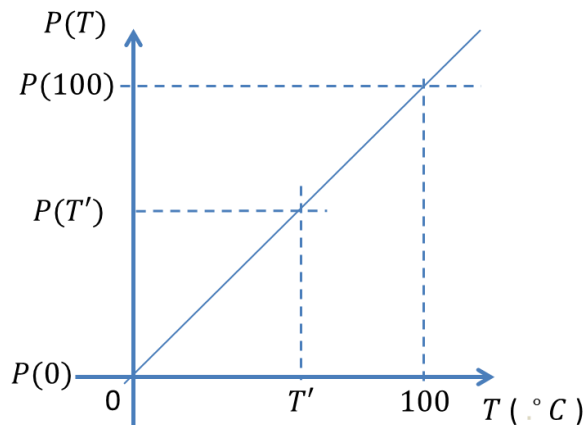
Thermometers and Temperature Scales (Pre-lecture Video)

Centigrade Scale:

It is based on the 1) ice point (0 °C) and the 2) steam point (100 °C). All other temperatures are determined by interpolation and extrapolation. For example, for the mercury-in-glass thermometer, the positions of the mercury is marked off at ice point and steam point. Then, the interval between these two marks is divided into one hundred equal marks. With this simplification, we can write

$$\frac{P(T') - P(0)}{T' - 0} = \frac{P(100) - P(0)}{100 - 0}$$

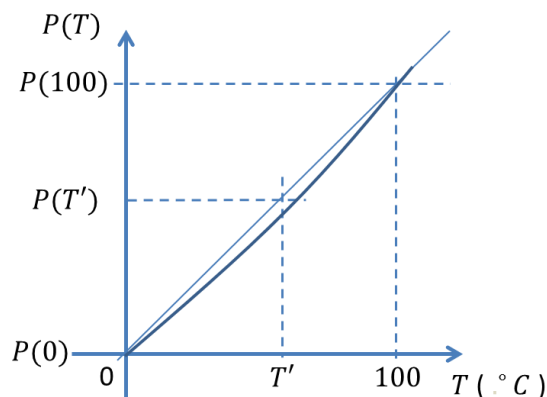
for some thermometric property $P(T)$ at temperature T (°C).



To find the corresponding temp T' for property $P(T')$, we write

$$T' = \frac{P(T') - P(0)}{P(100) - P(0)} \times 100$$

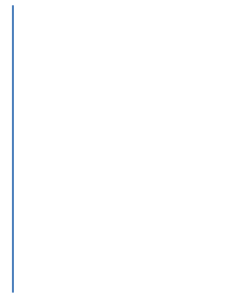
In reality, the mercury does not expand uniformly so there is a slight deviation from the correct temperature. This applies similarly to other thermometers.



Example [Muncaster, Ex 13.1] (Pre-lecture Video)

A particular resistance thermometer has a resistance of $30.00\ \Omega$ at ice point, $41.58\ \Omega$ at the steam point and $34.59\ \Omega$ when immersed in a boiling liquid. A constant gas thermometer gives readings of $1.333 \times 10^5\ \text{Pa}$, $1.821 \times 10^5\ \text{Pa}$ and $1.528 \times 10^5\ \text{Pa}$ at the same three temperatures. Calculate the temperature at which the liquid is boiling

- a) On the scale of the gas thermometer;
- b) On the scale of the resistance thermometer.



Ideal Gases

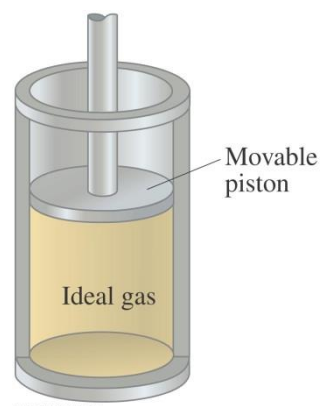
The Mole Concept

One mole of substance consist of $N_A = 6.022 \times 10^{23}$ elementary units (atoms or molecules depending on the substance). This is the same as number of atoms in 12 g of carbon-12. The number $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ is known as Avogadro's constant.

How many molecules are there is 15 g of O_2 gas? [1 mole of O_2 is 32 g, i.e. relative molecular mass, R.M.M. of O_2 is 32.]

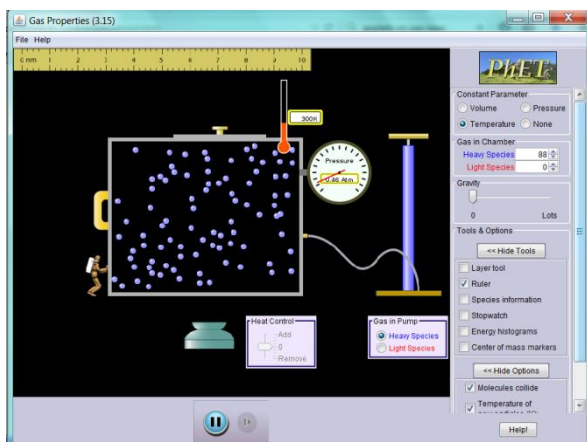
Gas Properties and Behaviour

What are some of the physical properties that we can measure for a fixed amount of gas confined in a container as shown on the right?



How do these physical properties relate to the collective microscopic behavior of gas atoms / molecules?

Gas Law



<http://phet.colorado.edu/en/simulation/gas-properties>

Try the simulation for yourself. For a fixed amount of gas at constant temperature T , the pressure p of the gas is inversely proportional to the volume V of the gas:

$$p \propto \frac{1}{V} \text{ (Boyle's Law)}$$

For a fixed amount of gas at constant pressure p , the volume V of the gas is proportional to the temperature T of the gas:

$$V \propto T \text{ (Charles' Law)}$$

For a fixed amount of gas at constant volume V , the pressure p of the gas is proportional to the temperature T of the gas:

$$p \propto T \text{ (Pressure Law)}$$

For a fixed n moles of ideal gas, its pressure p (Pa, N m^{-2}), volume V (m^3) and temperature T (K) follow the ideal gas equation

$$pV = nRT \text{ (Ideal Gas Law)}$$

where $R = 8.314 \text{ J/ (mol K)}$ is the gas constant.

Ideal Gas Equation

For a fixed n moles of ideal gas, its pressure p (Pa, N m^{-2}), volume V (m^3) and temperature T (K) follow the ideal gas equation

$$pV = nRT$$

where $R = 8.314 \text{ J / (mol K)}$ is the gas constant.

Remarks:

1. No real gases obey the ideal gas equation. However, the ideal gas equation is a good description for real gases at low densities, low pressures and high temperatures – microscopically, the gas molecules are far apart and their interactions negligible except during elastic collisions.

2. Some basic assumptions about behaviour of gas molecules in subsequent analysis:

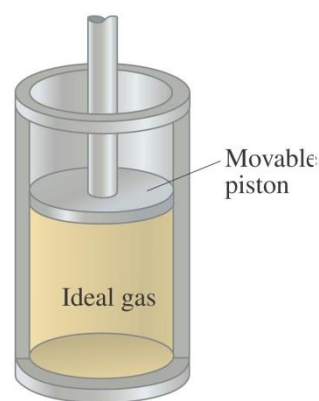
- large number of molecules, moving in random directions with a variety of speeds
- molecules are far apart, on average
- molecules obey laws of classical mechanics and interact only when colliding
- collisions are perfectly elastic
- the volume occupied by the gas molecules is small compared to the size of the container

3. Thus, we have the familiar gas laws :

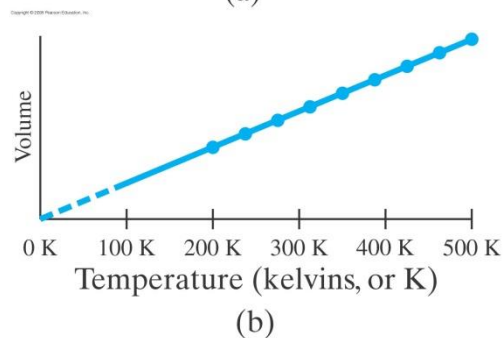
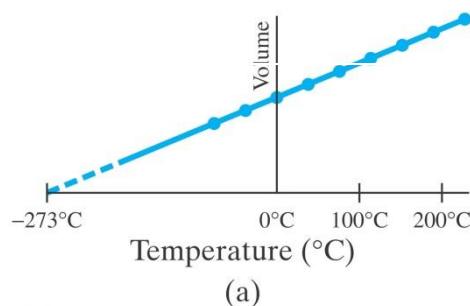
- $pV = \text{constant}$ [Boyle's Law];
- $V/T = \text{constant}$ [Charles' Law];
- $p/T = \text{constant}$ [Pressure Law]

4. This is a good point to introduce a new temperature scale, the kelvin scale. If we imagine that the temperature of the gas is reduced, the volume will also be reduced. So the question is, where the volume be equal to zero at some temperature if the gas does not liquefy /solidify?

5. $T/K = 273.15 + T/^{\circ}\text{C}$



Giancoli pg 508 Fig 19.7:
Gas in a Piston



Giancoli pg 464 Fig 17.14:
Celsius scale and Kelvin Scale –
where the number 273.15 comes from.

Thermometers and Temperature Scales [Video]

Constant volume gas thermometer; Kelvin Scale

For a fixed mass of gas at constant volume, we have

$$p = \left(\frac{nR}{V}\right) T.$$

This suggests that the pressure p of a gas behaving ideally is a good thermometric property. The relation is linear and does not depend on the properties of the selected gas. The operational mathematical statement representing a real gas behaving as an ideal gas is

$$\lim_{P_{tr} \rightarrow 0}$$

Kelvin Scale

The Kelvin scale uses the triple point of water (273.16 K) as the upper fixed point. The lower fixed point is defined to be at temperature zero (also known as absolute zero). Thus, where tr indicates the triple point of water

$$p_{tr} = \left(\frac{nR}{V}\right) T_{tr};$$

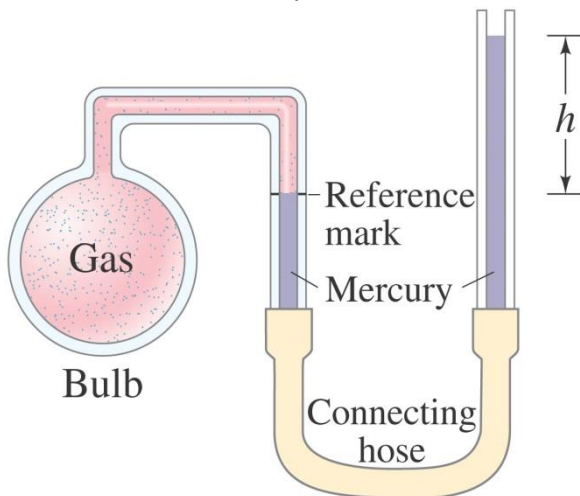
And for pressure p at an unknown temperature T' ,

$$p = \left(\frac{nR}{V}\right) T'.$$

Combining the equations, we can write

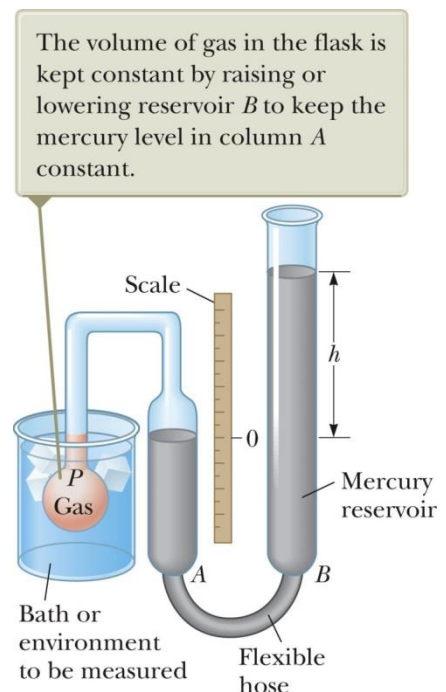
$$T = 273.16 \text{ K} \lim_{P_{tr} \rightarrow 0} \frac{P}{P_{tr}}$$

where $\lim_{P_{tr} \rightarrow 0}$ indicates that the real gases behaviour ideally in the low pressure limit. It also gives the principle guiding how the measurement should be taken and raw data should be interpreted.



Gas Thermometer

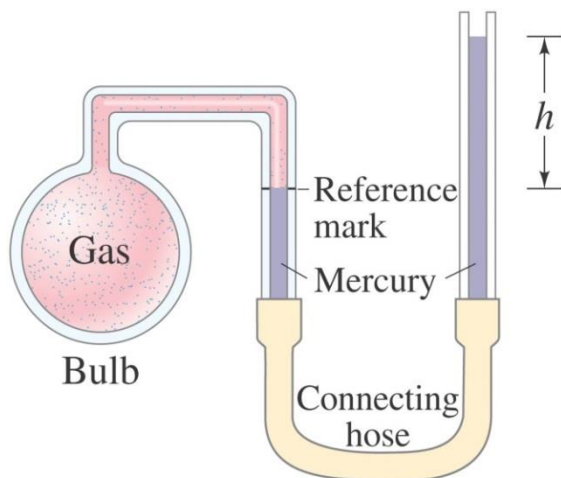
Giancoli Fig 13.1



Serway Fig 18.3

Constant Volume Gas Thermometer [Video]

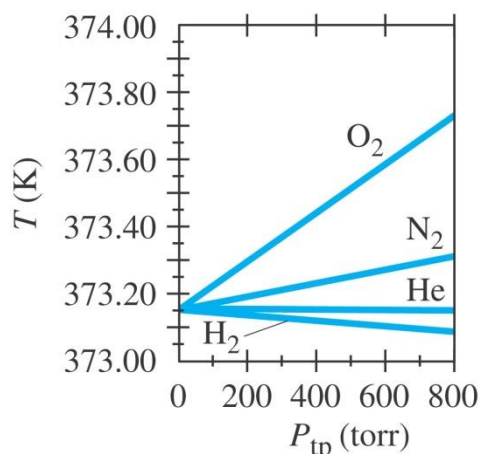
Essentially, we are applying Charles' law in the constant gas thermometer. This thermometer is used to set reference temperatures for other thermometers.



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Gas Thermometer

Giancoli Fig 13.1



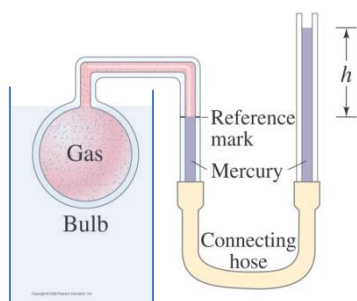
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Giancoli pg 470 Fig 17.17:

Measuring the temperature of boiling water using different gases and at diminishing pressures, $\lim_{P_{tr} \rightarrow 0}$. We note that at the number of moles of gas used is very small, the thermometers readings converge at one value, regardless of the type of gas used – they all behave close to ideal.

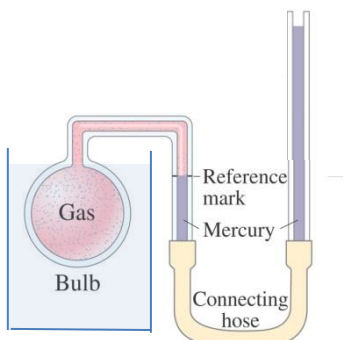
Young and Freedman Prob 17.8

A constant volume gas thermometer registers an absolute pressure corresponding to 325 mm of mercury when in contact with water at triple point temperature. What pressure does it read when placed in contact with water at normal boiling point?



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Water at boiling point



Water at 273.16 K,
triple point temperature



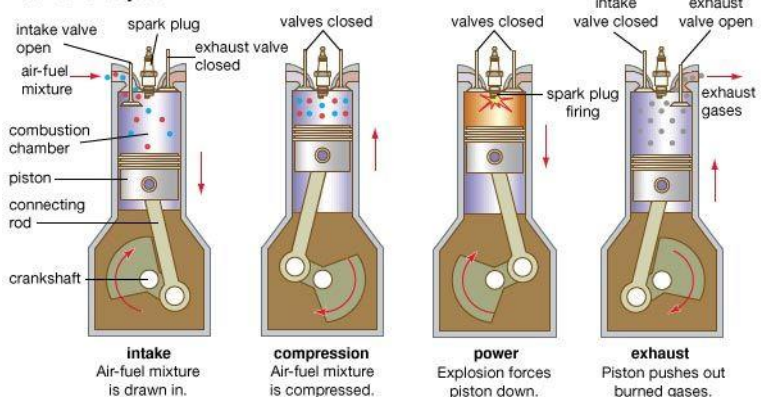
Giancoli Example 17-10

Determine the volume of 1.00 mol of any gas, assuming it behaves like an ideal gas, at STP (Standard temperature and Pressure: $T = 0^\circ\text{C}$, $P = 1\text{ atm} = 1.013 \times 10^5\text{ Pa}$).

Class Discussion: Giancoli Prob 17.32

In an internal combustion engine, air at atmospheric pressure and a temperature of about 20°C is compressed in the cylinder by a piston to $1/8$ of its original volume (compression ratio = 8.0). Estimate the temperature of the compressed air, assuming the pressure reaches 40 atm.

Four-stroke cycle



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<https://kids.britannica.com/kids/article/internal-combustion-engine/399511>

Class Discussion: Modified 2019 Exam Question 3 (i) – (iii), Thermodynamics

An idealized Stirling cycle shown in Figure 3 is made up of two isothermal processes and two isochoric process. Assume 0.008 moles of an ideal diatomic gas is used in this cycle. The temperature and volume of the gas at point 4 is $T_4 = 27^\circ\text{C}$ and $V_4 = 1.20 \times 10^{-4}\text{m}^3$ respectively. The temperature at point 1 is $T_1 = 457^\circ\text{C}$ and the volumes at points 2 and 3 are $V_2 = V_3 = 6.0 \times 10^{-4}\text{m}^3$.

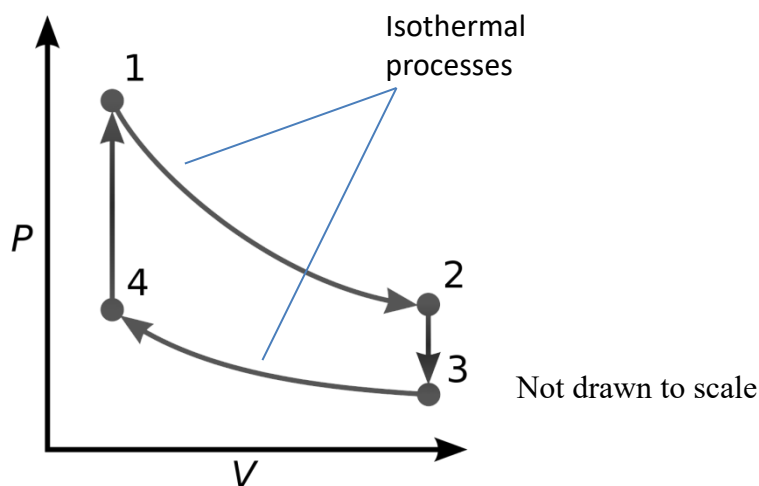


Figure 3

1. Determine the pressure P_4 at point 4.
2. Determine the change in *temperature* in the process $4 \rightarrow 1$.

The heat engine involves heat gain at constant volume and expansion at constant temperature. Identify these two processes in Figure 3.

Heat gain at constant volume: _____ $\rightarrow$ _____.

Expansion at constant temperature: _____ $\rightarrow$ _____.

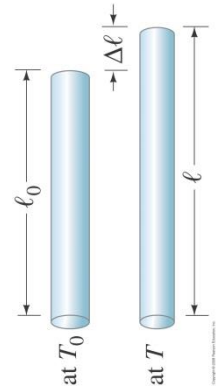
Thermal Expansion

Linear Expansion

When the temperature of a solid object changes, the change in length ΔL is (approximately) proportional to the change in temperature ΔT (if it is not too large):

$$\Delta L = \alpha L_0 \Delta T \text{ and } L = L_0 (1 + \alpha \Delta T)$$

where L_0 is the original length and α characterizes the length expansion of a particular material; it is called the coefficient of linear expansion and has units K^{-1} or $^{\circ}C^{-1}$.



Volume Expansion

When the temperature of an object changes, the change ΔV in its volume is (approximately) proportional to the temperature change ΔT . That is

$$\Delta V = \beta V_0 \Delta T \text{ and } V = V_0 (1 + \beta \Delta T)$$

where V_0 is the original volume and β characterizes the volume expansion of a particular material with units K^{-1} or $^{\circ}C^{-1}$. The quantity β is called the coefficient of volume expansion. Do you notice an interesting relation between α and β in the table below?

TABLE 17–1 Coefficients of Expansion, near $20^{\circ}C$

Material	Coefficient of Linear Expansion, α ($^{\circ}C$) $^{-1}$	Coefficient of Volume Expansion, β ($^{\circ}C$) $^{-1}$
<i>Solids</i>		
Aluminum	25×10^{-6}	75×10^{-6}
Brass	19×10^{-6}	56×10^{-6}
Copper	17×10^{-6}	50×10^{-6}
Gold	14×10^{-6}	42×10^{-6}
Iron or steel	12×10^{-6}	35×10^{-6}
Lead	29×10^{-6}	87×10^{-6}
Glass (Pyrex [®])	3×10^{-6}	9×10^{-6}
Glass (ordinary)	9×10^{-6}	27×10^{-6}
Quartz	0.4×10^{-6}	1×10^{-6}
Concrete and brick	$\approx 12 \times 10^{-6}$	$\approx 36 \times 10^{-6}$
Marble	$1.4\text{--}3.5 \times 10^{-6}$	$4\text{--}10 \times 10^{-6}$
<i>Liquids</i>		
Gasoline		950×10^{-6}
Mercury		180×10^{-6}
Ethyl alcohol		1100×10^{-6}
Glycerin		500×10^{-6}
Water		210×10^{-6}
<i>Gases</i>		
Air (and most other gases at atmospheric pressure)		3400×10^{-6}

Giancoli pg 460
Table 17.1

Giancoli Prob 17.8

A concrete highway is built of slabs 12 m long (15°C). How wide should the expansion cracks between the slabs be (at 15°C) to prevent buckling if the range of temperature is -30°C to 50°C?

Explain $\beta \approx 3\alpha$

Consider a rectangular solid of length l_o , width w_o , and height h_o . If the temperature changes by ΔT , its volume changes from $V_o = l_o w_o h_o$ to

$$V = l_o(1 + \alpha\Delta T)w_o(1 + \alpha\Delta T)h_o(1 + \alpha\Delta T).$$

Thus,

$$\Delta V = V - V_o = V_o(1 + \alpha\Delta T)^3 - V_o = V_o[3\alpha\Delta T + 3(\alpha\Delta T)^2 + (\alpha\Delta T)^3]$$

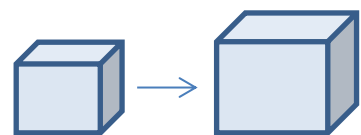
$$\Delta V \approx (3\alpha)V_o\Delta T$$

To appreciate the approximation, use a numerical value say $\alpha\Delta T = 0.02$. We note that higher power terms of $\alpha\Delta T$ are negligible.

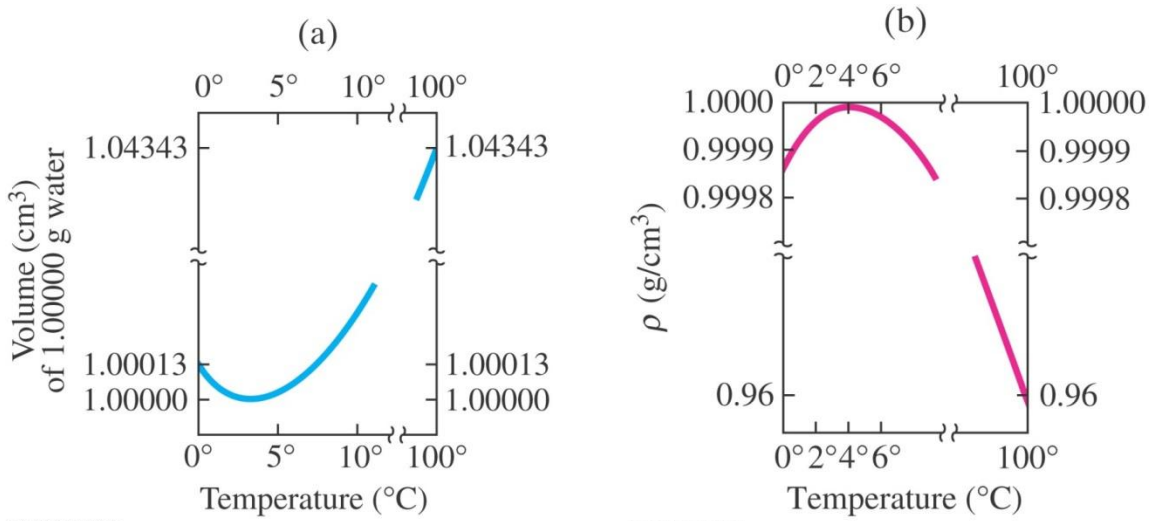
Giancoli Example 17-7: Gas Tank in Sun.

The 70-liter (L) steel gas tank of a car is filled to the top with gasoline at 20°C. The car sits in the Sun and the tank reaches a temperature of 40°C. How much gasoline do you expect to overflow from the tank?

$$\alpha_{steel} = 12 \times 10^{-6} (\text{ }^\circ\text{C})^{-1}; \beta_{gasoline} = 950 \times 10^{-6} (\text{ }^\circ\text{C})^{-1}$$



Anomalous behaviour of water below 4 °C



Giancoli Fig 17.12

Most substances contract more or less uniformly with temperature decrease as long as there is no phase change. As seen above, water expands when temperature is reduced from 4°C to 0°C.



<https://www.youtube.com/watch?v=UukRgqzk-KE&t=5s>

Heat Capacities and Latent Heat



Aluminum



Wood

Different amounts of thermal energy is required to raise the temperature of same mass substance by 1 °C or 1 K.



The amount of thermal energy required to raise the temperature of the same substance by 1 °C or 1 K is dependent on the mass.

TABLE 19–1 Specific Heats
(at 1 atm constant pressure and 20°C unless otherwise stated)

Substance	Specific Heat, c	
	kcal/kg · °C (= cal/g · °C)	J/kg · °C
Aluminum	0.22	900
Alcohol (ethyl)	0.58	2400
Copper	0.093	390
Glass	0.20	840
Iron or steel	0.11	450
Lead	0.031	130
Marble	0.21	860
Mercury	0.033	140
Silver	0.056	230
Wood	0.4	1700
Water		
Ice (–5°C)	0.50	2100
Liquid (15°C)	1.00	4186
Steam (110°C)	0.48	2010
Human body (average)	0.83	3470
Protein	0.4	1700

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1. How much energy is needed to raise 3.0 kg of Al by 4°C?

2. How much will the temperature of a piece of 2.0 kg Al change if 1800 J of energy is removed from it?

Heat Capacities and Latent Heat

The heat capacity (C) of a body is defined as being the heat required to produce a unit temperature change. Units: J K^{-1} or $\text{J }^{\circ}\text{C}^{-1}$.

$$\Delta Q = C \Delta T$$

The heat capacity per unit mass of a substance is the specific heat capacity (c). It is characteristic of the substance of which the body is composed.

Units: $\text{J kg}^{-1} \text{K}^{-1}$ or $\text{J kg}^{-1} ^{\circ}\text{C}^{-1}$.

$$\Delta Q = mc \Delta T ; C = mc$$

Since we are dealing only with temperature changes, the numeric values of C when expressed in J K^{-1} is the same as those for $\text{J }^{\circ}\text{C}^{-1}$. The same applies to c .

The specific latent heat of (l) of fusion (or vaporization, or sublimation) of a substance is the energy required to cause unit mass of the substance to change between solid and liquid (or liquid and vapour, or solid and vapour) without temperature change. Units: J kg^{-1} .

$$\Delta Q = ml$$

TABLE 19–2 Latent Heats (at 1 atm)

Substance	Melting Point ($^{\circ}\text{C}$)	Heat of Fusion		Boiling Point ($^{\circ}\text{C}$)	Heat of Vaporization	
		kcal/kg [†]	kJ/kg		kcal/kg [†]	kJ/kg
Oxygen	−218.8	3.3	14	−183	51	210
Nitrogen	−210.0	6.1	26	−195.8	48	200
Ethyl alcohol	−114	25	104	78	204	850
Ammonia	−77.8	8.0	33	−33.4	33	137
Water	0	79.7	333	100	539	2260
Lead	327	5.9	25	1750	208	870
Silver	961	21	88	2193	558	2300
Iron	1808	69.1	289	3023	1520	6340
Tungsten	3410	44	184	5900	1150	4800

[†]Numerical values in kcal/kg are the same in cal/g.

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Example: How much energy is needed to melt 3 kg of pure ice at -10°C ?

Giancoli Prob 19.21

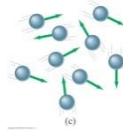
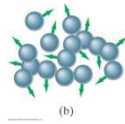
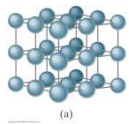
High-altitude mountain climbers do not eat snow, but always melt it first with a stove. To see why, calculate the energy absorbed from your body if you (a) eat 1.0 kg of -10°C snow which your body warms to body temperature of 37°C . (b) You melt 1.0 kg of -10°C snow using a stove and drink the resulting 1.0 kg of water at 2°C , which your body has to warm to 37°C .

Example: Circuit melt down [Young (College) pg 452, Ex 14.7]

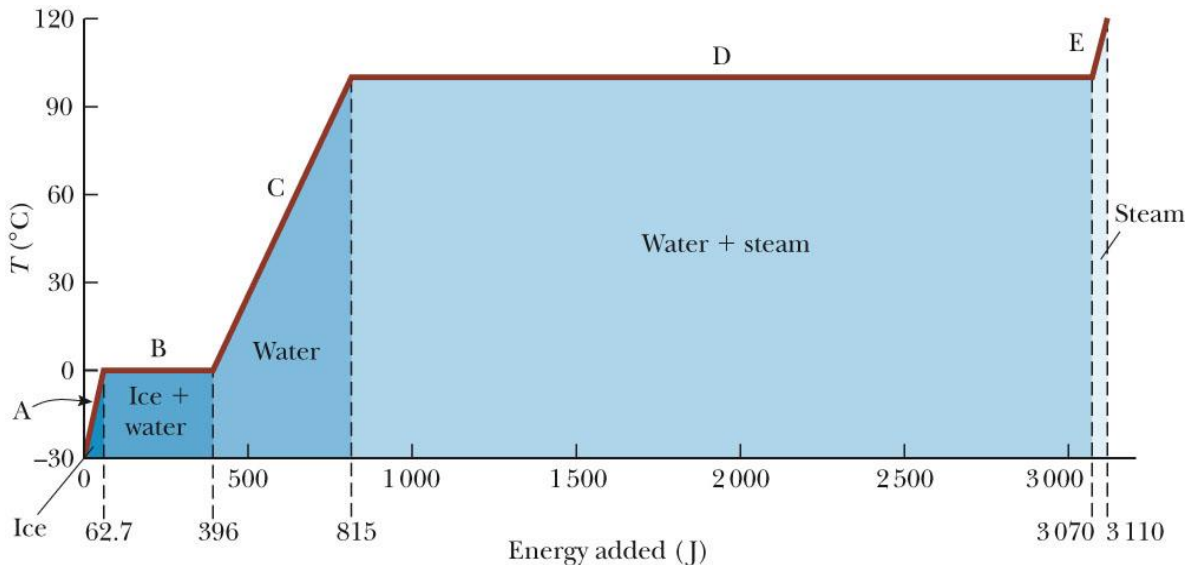
You are designing an electronic circuit element made of 23 mg of silicon. The electric current through it adds energy at the rate $7.4 \times 10^{-3} \text{ W}$. Specific heat of silicon is $705 \text{ J}/(\text{kg K})$. Assume no heat lost to surrounding. What is rate of temperature rise?

In one second,

Plot of temp vs energy added to 1.0 g of water from -30°C to 120°C



Giancoli pg 456 Fig 17.2:
The atomic picture of solid,
liquid and gaseous phase



From Jewett & Serway Vol 1 Chap 20; Fig 20.3, pg 574

An analysis of heat supplied to 1.0 g of water from -30°C to 120°C

Part A: Ice (-30.0°C to 0.0°C) ; $c_{\text{ice}} = 2\,090\text{ J} / (\text{kg } ^\circ\text{C}^{-1})$. [Warming up]

$$Q_1 = 62.7\text{ J}$$

Part B: Ice $\rightarrow$ Water (0.0°C); $L_{f, \text{ice}} = 333\,000\text{ J} / \text{kg}$.
[Change of state from solid to liquid]

$$Q_2 = 330\text{ J}$$

Part C: Water (0.0°C to 100.0°C) ; $c_{\text{water}} = 4\,190\text{ J} / (\text{kg } ^\circ\text{C}^{-1})$. [Warming up]

$$Q_3 = 419\text{ J}$$

Part D: Water $\rightarrow$ Steam (100.0°C); $L_{v, \text{water}} = 2\,260\,000\text{ J} / \text{kg}$.
[Change of state from liquid to gas]

$$Q_4 = m_{\text{water}} l_{\text{water}} = (1.0 \times 10^{-3} \text{ kg}) \left(22.6 \times 10^5 \frac{\text{J}}{\text{kg}} \right) = 2260\text{ J}$$

Part E: Steam (100.0°C to 120.0°C) ; $c_{\text{steam}} = 2010\text{ J} / (\text{kg } ^\circ\text{C}^{-1})$. [Warming up]

$$Q_5 = m_{\text{water}} c_{\text{water}} \Delta T = (1.0 \times 10^{-3} \text{ kg}) \left(2010 \frac{\text{J}}{(\text{kg } ^\circ\text{C})} \right) ((120 - 100)^\circ\text{C}) = 40.2\text{ J}$$

Giancoli Prob 19.12

A hot iron horseshoe of mass 0.40 kg just forged, is dropped into 1.05 L of water in a 0.30-kg iron pot initially at 20.0°C. If the final equilibrium temperature is 25.0°C, estimate the initial temperature of the hot horseshoe.



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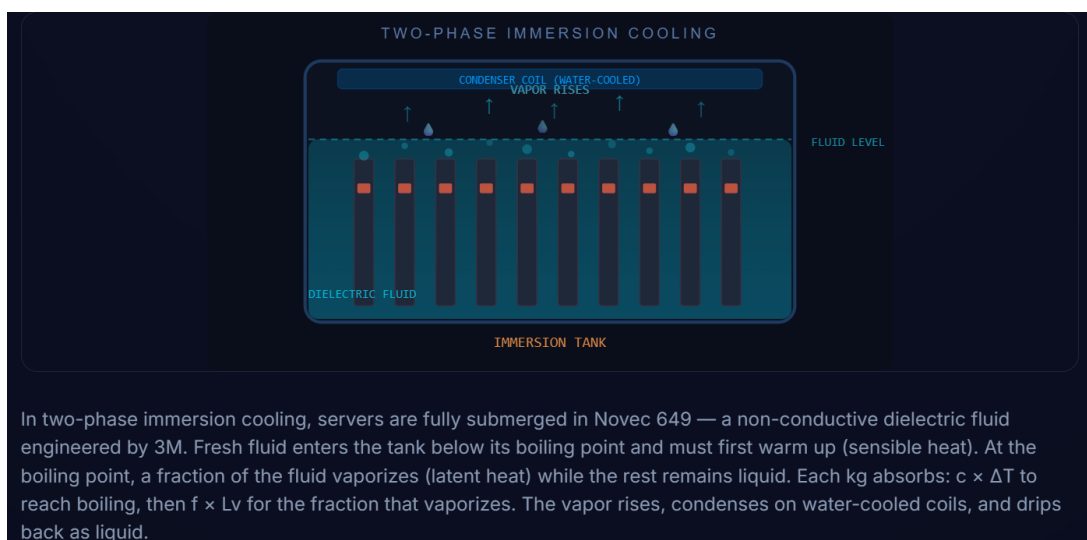
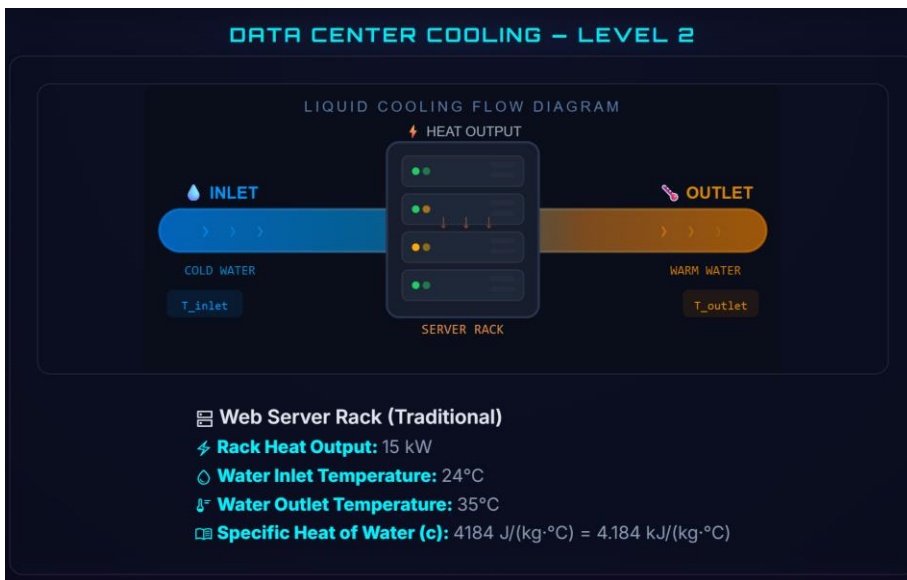


<https://www.youtube.com/watch?v=lykTK45bXrl&t=5s>

Gemini 2.5 Pro Generated

A 0.50 kg lead weight, initially at 100.0°C, is dropped into a 0.20 kg aluminium calorimeter containing 0.40 L of water. The calorimeter and water are both initially at 20.0°C. What is the final equilibrium temperature of the system? [Assume no heat is lost to the surroundings and the density of water is 1.0 kg/L. $c_{Pb} = 128 \text{ J}(\text{kg} \cdot \text{K})^{-1}$, $c_{Al} = 900 \text{ J}(\text{kg} \cdot \text{K})^{-1}$, $c_{water} = 4186 \text{ J}(\text{kg} \cdot \text{K})^{-1}$]

Class Discussion: Data Center Cooling



Published on 27 Oct 2025

Proposal for carbon-neutral data centres in space



An out-of-this-world idea: placing data centres in space could pave the way for sustainable computing with unlimited solar energy and free cooling, says scientists from **NTU Singapore**.

The researchers outline a practical path to building carbon-neutral data centres in Low Earth Orbit (LEO), a concept particularly relevant to land-scarce cities like Singapore, where limited land and high real estate costs make conventional data centres increasingly expensive.

Published in the peer-reviewed journal *Nature Electronics*, the study presents a framework for how satellites equipped with advanced processors could serve as orbital edge and cloud data centres.

Which argument do you support?

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FEATURE COMPUTING

WANT AN ENERGY-EFFICIENT DATA CENTER? BUILD IT UNDERWATER

Microsoft wants to submerge data centers to keep them cool and to harvest energy from the sea

<https://spectrum.ieee.org/want-an-energyefficient-data-center-build-it-underwater/particle-1>

versus

06 NOV 2018



MICROSOFT PROJECT NATICK DATA CENTRE RENEWABLE ENERGY AZURE

Heat waste of Microsoft's submarine data centre not acceptable

By Ashton Young

Last week at the Microsoft Future Decoded conference in London, Satya Nadella shared with the audience that he believes *underwater data centres are the way of the future*.

Of course, he's referring to the container-sized submarine data centre (Project Natick) that holds 12 racks containing 864 servers, all while using a liquid cooling system that transfers heat to the surrounding ocean.

Microsoft launched the revolutionary data centre in June this year, which is now processing workloads on the seafloor near Scotland's Orkney Islands.

"Since 50 percent of the world's population lives close to water bodies, we think this is the way we want to think about future data center regions and expansion," Nadella said.

<https://datacenternews.asia/story/heat-waste-of-microsoft-s-submarine-data-centre-not-acceptable>

More Facts and Opinons

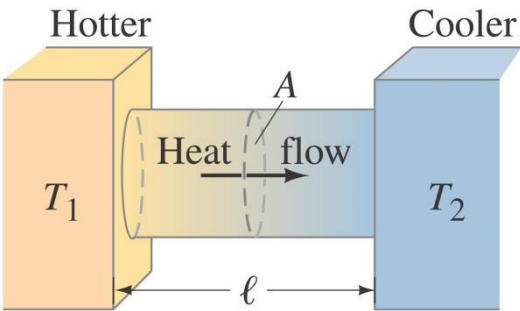
<https://energyinnovation.org/2020/03/17/how-much-energy-do-data-centers-really-use/>

<https://www.carbonbrief.org/factcheck-what-is-the-carbon-footprint-of-streaming-video-on-netflix?>

Heat Transfer: Conduction, Convection and Radiation

(extracted from https://en.wikipedia.org/wiki/Heat_transfer)

Conduction is the transfer of energy between objects that are in physical contact. On a microscopic scale, heat conduction occurs as hot, rapidly moving or vibrating atoms and molecules interact with neighbouring atoms and molecules, transferring some of their energy (heat) to these neighbouring particles. In other words, heat is transferred by conduction when adjacent atoms vibrate against one another, or as electrons move from one atom to another. Conduction is the most significant means of heat transfer within a solid or between solid objects in [thermal contact](#). Fluids—especially gases—are less conductive.



Assuming no heat lost to surrounding.

$$\frac{\Delta Q}{\Delta t} = kA \frac{T_1 - T_2}{\ell}.$$

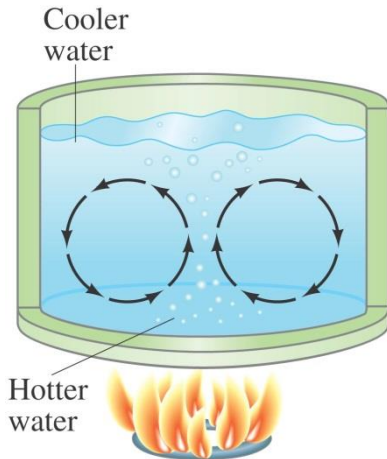
TABLE 19–5
Thermal Conductivities

Substance	Thermal conductivity, <i>k</i>	
	kcal (s · m · C°)	J (s · m · C°)
Silver	10 × 10 ^{−2}	420
Copper	9.2 × 10 ^{−2}	380
Aluminum	5.0 × 10 ^{−2}	200
Steel	1.1 × 10 ^{−2}	40
Ice	5 × 10 ^{−4}	2
Glass	2.0 × 10 ^{−4}	0.84
Brick	2.0 × 10 ^{−4}	0.84
Concrete	2.0 × 10 ^{−4}	0.84
Water	1.4 × 10 ^{−4}	0.56
Human tissue	0.5 × 10 ^{−4}	0.2
Wood	0.3 × 10 ^{−4}	0.1
Fiberglass	0.12 × 10 ^{−4}	0.048
Cork	0.1 × 10 ^{−4}	0.042
Wool	0.1 × 10 ^{−4}	0.040
Goose down	0.06 × 10 ^{−4}	0.025
Polyurethane	0.06 × 10 ^{−4}	0.024
Air	0.055 × 10 ^{−4}	0.023

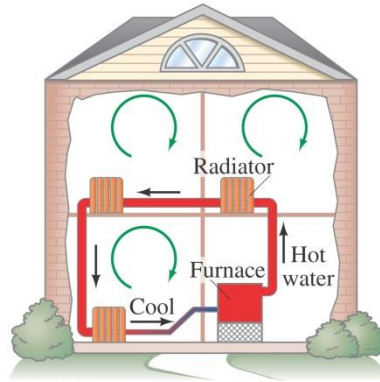
Heat Transfer: Conduction, Convection and Radiation

(extracted from https://en.wikipedia.org/wiki/Heat_transfer)

Convection occurs when heat flows by the mass movement of molecules from one place to another. It may be natural or forced; both these examples are natural convection.



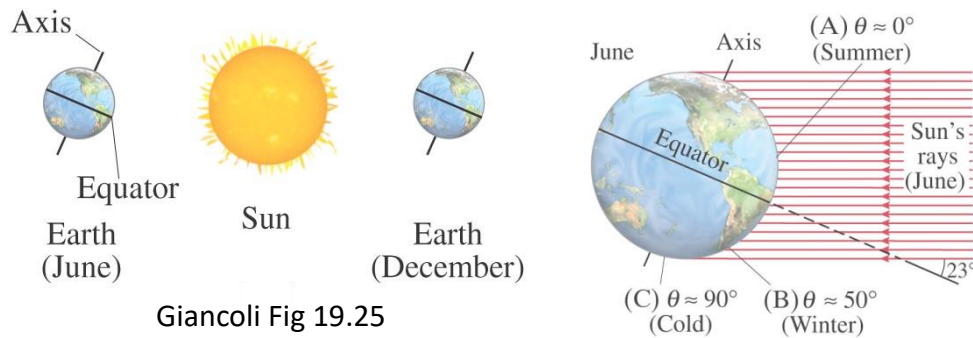
Giancoli Fig 19.22



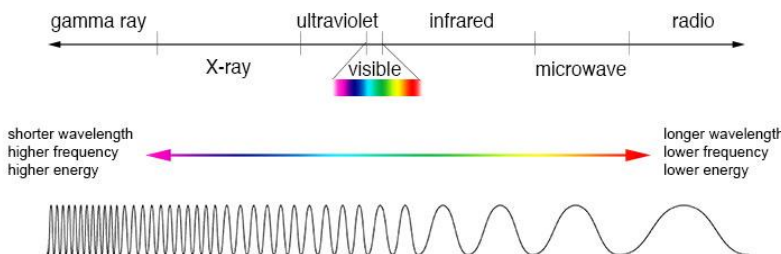
Giancoli Fig 19.23

In the above diagrams, convection currents are set up as the heated fluid at the bottom rises because of reduced density and the heated fluid is replaced by cooler fluid from above.

Thermal radiation occurs through a vacuum or any transparent medium (solid or fluid). It is the transfer of energy by means of electromagnetic waves.



Giancoli Fig 19.25



<https://imagine.gsfc.nasa.gov/science/toolbox/emsp/ectrum1.html>